

1.5 To-be state SLAs

1.5.1 SLA measurement and reporting

- i. The SLA monitoring and reporting platform should be able to measure and report all the defined SLAs as per their corresponding measurement frequency and reporting intervals. The SLA definition also includes the measurement methodology and process to be adopted for measurement including the sampling interval for each SLA.
- ii. The MSIP shall design and implement appropriate scripts/code to perform the required measurements and the same should be integrated with the SLA dashboard.
- iii. The SLA dashboard should show the SLA metrics in tabular and graphical formats for all monitored resources and services. The dashboard shall also display the downtime, uptime or any other parameter for the services and service groups along with compliance or deviations from the SLA. The reports shall be available for download in CSV and PDF format for any given date and time range along with daily, weekly, monthly and yearly reports. The MSIP shall maintain the SLA data online for the entire duration of the contract.
- iv. The SLA dashboard should be developed using open source tools, products such as Grafana, Chronograf, etc.
- v. For measuring application throughput and response time, the MSIP shall provide a logging and analysis infrastructure along with a customizable dashboard to measure, monitor and track the application SLAs. Tools like Logstash, Elasticsearch and Kibana should be integrated with the SLA dashboard. The MSIP shall co-ordinate with the MSAP, BSP, SDAs or other users of the infrastructure to implement the application SLA measurement and monitoring.
- vi. The MSIP shall provide a fully automated dashboard of all SLAs with alerts indicating, compliance or deviation from the defined thresholds.
- vii. The SLA measurement and reporting system and the EMS should be highly available so that it has minimal or near zero down time. The MSIP is expected to design the EMS and SLA Management infrastructure accordingly.
- viii. For computation of availability SLAs, downtime of server, storage, network and security components supplied by MSIP due to data center environment failures, i.e., power and HVAC failures, shall be excluded.

1.5.2 SLA Classification

The To-be state SLAs have been classified into two categories:

- Milestone based SLAs and
- Operational SLAs which measure various aspects such as availability, performance, reliability, provisioning time, and quality of managed services.

Each of these SLAs is described in the following sections.

1.5.3 Milestone based SLAs

- i. The table below shows the various activities and timelines which are to be treated as key project milestones and the Severity levels corresponding to them.
- ii. The MSIP shall use Project Management Tool (to be procured and configured by MSAP) to continuously track and monitor the project progress along with dependencies.
- iii. The MSIP shall submit a weekly status report along with the milestone report in the Project Management Tool.



- iv. The SLAs linked to milestones are applied once on delay beyond the specified target timeline.



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Table 1: Milestone based SLAs

S. No.	Category	Description	Target for completion (T being date of signing of contract)	Reporting tools to be used	Reports/data to be submitted	Severity level for application of liquidated damages
1	Initial team deployment – First Milestone	Deployment of 30% of staff (including all key profiles as per clause 8.11 of Vol I)	T+4 weeks	Project Management Tool	Deployment Completion Report	2
2	Team deployment – Third Milestone	Deployment of 100% of staff	T+3 months	Project Management Tool	Deployment Completion Report	2
3	Transition and takeover of as-is infrastructure and as-is SLAs	Completion of takeover of as-is CIDR IT operations including NOC and SOC and commencement of As-Is Infrastructure related SLAs	T+ 3 months	Project Management Tool	Transition Completion Report	3
4	Finalize detailed design and Transition Plan	Finalize detailed design and plan to transition from As-Is Infrastructure to To-Be state Infrastructure	T+ 3 months	Project Management Tool	Detailed design and Transition Plan	2
5	EMS Tools – First Release	Commissioning and Go-live of EMS tools for - Incident Management - Service Management - Defect management and - Trouble Ticketing - IT Asset Management	T+ 3 months	NA	UAT Test Plan for EMS Tools – First release	3



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S. No.	Category	Description	Target for completion (T being date of signing of contract)	Reporting tools to be used	Reports/data to be submitted	Severity level for application of liquidated damages
		Note: Since the To-Be IT infrastructure will not be ready by this time, MSIP can deploy these tools on the As-Is IT infrastructure to start with. This milestone SLA applies to deployment of the tools on As-Is IT infrastructure. MSIP can subsequently migrate these tools to the To-Be IT infrastructure.				
6	Site Readiness assessment	MSIP to assess the rack space in both data centres, carry out necessary measurements and planning for cabling and deployment and submit Site Readiness Assessment Report	T+3 months	Project Management Tool	Site readiness assessment report of HDC and MDC confirming readiness of the site for hardware deployment.	2
6.A	Proof of Concept	Fully functional POC as defined in clause 6.4 (4) of Volume-II on existing hardware	T+6 months	NA	MSIP should provide demonstrations to UIDAI of all provisioning capabilities	1



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S. No.	Category	Description	Target for completion (T being date of signing of contract)	Reporting tools to be used	Reports/data to be submitted	Severity level for application of liquidated damages
7	Delivery of IT assets	Delivery of IT assets for To-be Infrastructure - Bengaluru DC	T+6 months	IT Asset Manager	Updated details in IT Asset Manager	3
8	Delivery of IT assets	Delivery of IT assets for To-be Infrastructure - Manesar DC	T+6 months	IT Asset Manager	Updated details in IT Asset Manager	3
9	Commissioning and Go-Live	Power on and Confirmation of readiness for UAT by MSIP of Physical IT infrastructure in Bengaluru DC. The following should be completed at this milestone: deployment of racks, servers, network and storage components, network cabling, deployment of iPDUs and power connections, OS installation, server hardening, setting up of network backbone and assignment of IP addresses to all components.	T+8 months or 5 months from the date of completion of Site Readiness Assessment (Milestone 6) for Bengaluru Data centre, whichever is later	Project Management Tool	UAT Test Plan and Acceptance Testing Report	3
10	Commissioning and Go-Live	Power on and Confirmation of readiness for UAT by MSIP of Physical IT infrastructure in Manesar DC.	T+8 months or 5 months from the date of completion of	Project Management Tool	UAT Test Plan and Acceptance Testing Report	3



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S. No.	Category	Description	Target for completion (T being date of signing of contract)	Reporting tools to be used	Reports/data to be submitted	Severity level for application of liquidated damages
		The following should be completed at this milestone: deployment of racks, servers, network and storage components, network cabling, deployment of iPDUs and power connections, OS installation, server hardening, setting up of network backbone and assignment of IP addresses to all components.	Site Readiness Assessment (Milestone 6) for Manesar Data centre, whichever is later			
Note: The date (whichever is later) on which both milestones 9 and 10 above are achieved will be treated as T2 and the same is referred to in the following milestones.						
11	Commissioning and Go-Live	Deployment of CRM and CCM solution in production	T + 12 months or T2+4 months, which is later [monthly design review after T+3 months]	Project Management Tool	UAT Test Plan and Acceptance Testing Report	3
12	Commissioning and Go-Live	Deployment of billing solution in production	T + 12 months or T2 + 4 months,	Project Management Tool	UAT Test Plan and Acceptance Testing Report	3



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S. No.	Category	Description	Target for completion (T being date of signing of contract)	Reporting tools to be used	Reports/data to be submitted	Severity level for application of liquidated damages
			whichever is later. [monthly design review after T+3 months]			
13	EMS Tools- Second Release	Deployment and confirmation of readiness for UAT of EMS Solution's SLA Management, Monitoring and Reporting module A web based module using open source technologies with dashboard for real time monitoring of all the defined SLAs, deviations from targets, if any and with provision for automated report generation and alerting. This module should also automatically compute the percentage of liquidated damages applicable at the end of each measurement interval and reporting interval.	T+ 12 months or T2 + 4 months, whichever is later. [monthly design review after T+3 months]	Project Management Tool	UAT Test Plan and Acceptance Testing Report	4



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S. No.	Category	Description	Target for completion (T being date of signing of contract)	Reporting tools to be used	Reports/data to be submitted	Severity level for application of liquidated damages
14	Commissioning and Go-Live	Deployment and provisioning of CI/CD Tools and confirmation of readiness for UAT in <i>Development, Testing, Staging & Pre-Production, Benchmarking and Production environments</i>	T+ 12 months or T2 + 4 months, whichever is later. [monthly design review after T+3 months]	Project Management Tool	UAT Test Plan and Acceptance Testing Report	3
15	Virtual Cloud Infrastructure	Deployment and confirmation of readiness for UAT of Cloud Infrastructure for Compute, Storage, Network in <i>Development, Testing, Staging & Pre-Production, and Benchmarking environments</i>	T+ 12 months or T2 + 4 months, whichever is later. [monthly design review after T+3 months]	Project Management Tool	UAT Test Plan and Acceptance Testing Report	3
16	Virtual Cloud Infrastructure	Deployment and confirmation of readiness for UAT of Cloud Infrastructure for Compute, Storage, Network in <i>Production environment</i>	T+ 14 months or T2 + 6 months, whichever is later.	Project Management Tool	UAT Test Plan and Acceptance Testing Report	3



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S. No.	Category	Description	Target for completion (T being date of signing of contract)	Reporting tools to be used	Reports/data to be submitted	Severity level for application of liquidated damages
			[monthly design review after T+3 months]			
17	IaaS Services	Deployment and confirmation of readiness for UAT of automated, integrated web based tool for provisioning and managing all IaaS Services across all environments	T+ 14 months or T2 + 6 months, whichever is later. [monthly design review after T+3 months]	Project Management Tool	UAT Test Plan and Acceptance Testing Report	3
18	PaaS Services	Deployment and confirmation of readiness for UAT of PaaS Services with complete automation across all environments - All PaaS Services specified in RFP should be available as a Web Based Service and accessible through a unified UIDAI private cloud portal	T+14 months or T2 + 6 months, whichever is later. [monthly design review	Project Management Tool	UAT Test Plan and Acceptance Testing Report	3



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S. No.	Category	Description	Target for completion (T being date of signing of contract) after T+3 months]	Reporting tools to be used	Reports/data to be submitted	Severity level for application of liquidated damages
19	EMS Tools-Third Release	Deployment and confirmation of readiness for UAT of EMS tools for complete end to end management, provisioning of all PaaS, IaaS Services and all infrastructure components and all ITIL Services on cloud infrastructure across all environments. All EMS Tools deployed on As-Is IT infrastructure as part of EMS Tools – First Release should be migrated to To-Be IT infrastructure as part of this release.	T+ 16 months or T2 + 8 months, whichever is later. [monthly design review after T+3 months]	Project Management Tool	UAT Test Plan and Acceptance Testing Report	4
20	EMS Tools-Fourth Release	Full implementation and confirmation of readiness for UAT of all EMS requirements specified in the RFP	T+18 months or T2 + 10 months, whichever is later. [monthly design review after T+3 months]	Project Management Tool	UAT Test Plan and Acceptance Testing Report	4



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S. No.	Category	Description	Target for completion (T being date of signing of contract)	Reporting tools to be used	Reports/data to be submitted	Severity level for application of liquidated damages
21	Data migration	Completion of storage migration of all existing data from current technology platform to new technology platform	T1+ 18 months where T1 is the date of completion of on-boarding of BSP or notification from UIDAI to MSIP to commence data migration	Project Management Tool	Data migration report confirming successful migration of data from existing technology platform to new technology platform	3
22	On-boarding of new TSPs	On-boarding of new TSPs and cutover from existing TSPs to new TSPs	T+3 months	Project Management Tool	NA	4

Note: In case of a delay in completion of milestone is attributable to UIDAI, such delay will be adjusted from the subsequent milestones.

1.5.4 Availability SLAs

1.5.4.1 To-Be SLA 01 - Availability of Physical Server or Virtual Machine, Container or PaaS Service

SLA Description	Availability of Physical Server or Virtual Machine, Container or PaaS Service
Scope of SLA	Applies to physical server, virtual machine, container or PaaS service
Tools, methods or scripts to be used to measure SLA	EMS tools to be implemented by MSIP – a custom monitoring script shall be executed once every minute from the SLA monitoring server.
Process to capture data or measure the parameter	In order to measure availability, it is not just sufficient to have the system is up and running. The MSIP can either use an agentless mechanism using ssh from the remote management servers or use an agent based solution for measuring the SLAs. <u>Measurement for Physical Server or VM</u> The availability of a physical server or VM shall be measured as follows. From the SLA monitoring server(s) connect via SSH to the server, VM under measurement and execute a script which checks, validates the following: - CPU Utilization should be less than 95%, Memory Utilization should be less than 90%, Free disk space should be more than 10%, and there should be no swapping happening. - The system should have sufficient system resources like – files, sockets, semaphores, open files, locks - All file systems defined in the fstab should be mounted and available with the defined permissions - The system should be able to connect to all external servers which it needs to access (or is accessed) via http(s) or TCP/IP. - The system services like DNS, DHCP Client, Firewall, HIDS, Antivirus and any other required services or agents should be up and running. - There shall be no critical or major errors in the system log files The SSH script should return a success if all the above checks are successful otherwise it is deemed as a failure. The Server or VM is deemed to have a downtime if either the SSH times out or returns a failure. The down time is measured from the last failed SSH script to the next successful SSH script execution. <u>Measurement for standalone PaaS service</u> The availability of a standalone PaaS Service shall be measured as follows: - From the SLA monitoring server(s) connect via JDBC/HTTP(s)/TCP request to invoke the service under measurement. - The service should return successfully within the baselined response time - The response time cannot be more than 120% of the baselined response time - The response time measurements will be applicable for 99 percentiles of the requests

SLA Description	Availability of Physical Server or Virtual Machine, Container or PaaS Service
	The monitoring script shall be executed once every minute from the SLA monitoring server. The PaaS Service is deemed to have a downtime if either the service request times out or returns a failure. The down time is measured from the last failed service request script to the next successful script execution. <u>Standalone container availability measurement</u> The MSIP shall deploy all container images with the HEALTHCHECK option enabled. Below are the steps to check availability. - The health check command should be executed every 60 seconds with a timeout period of 3 seconds from the management server. - The container is down if the health check command returns UNHEALTHY. - The response time measurements will be applicable for 99 percentiles of the requests The downtime is measured from the last failed UNHEALTHY Command to the next successful HEALTHCHECK command execution.
SLA calculation	Will be calculated separately for physical machines, virtual machines, containers and PaaS services using the below formulae. Formula – 1: Availability % for Physical Machines = $\{1 - [(Total\ Downtime - Planned\ downtime) / (Total\ Time - Planned\ Downtime)]\} * 100$ <i>Total cumulative downtime</i> $= \sum_{1}^n \text{downtime of all physical servers during the measurement interval}$ [For example, if there are 600 physical servers, the downtimes recorded automatically by the monitoring script as per the process defined in “Measurement for Physical Server or VM” above will be summed up (i.e., cumulative downtime will be computed)] Planned Downtime for physical servers = 1 hour per quarter Total Time = 24 X 7 measured over a period of one calendar month. Formula - 2: Availability % for Virtual Machines = $\{1 - [(Total\ Downtime - Planned\ downtime) / (Total\ Time - Planned\ Downtime)]\} * 100$

SLA Description	Availability of Physical Server or Virtual Machine, Container or PaaS Service
	<i>Total cumulative downtime</i> $= \sum_{1}^n \text{downtime of all virtual machines during the measurement interval}$ [For example, if there are 600 virtual machines, the downtimes recorded automatically by the monitoring script as per the process defined in "Measurement for Physical Server or VM" above will be summed up (i.e., cumulative downtime will be computed)] Planned Downtime for virtual machines = 1 hour per quarter Total Time = 24 X 7 measured over a period of one calendar month. Formula - 3: Availability % for PaaS services = {1- [(Total Downtime-Planned downtime) / (Total Time – Planned Downtime)]} *100 <i>Total cumulative downtime</i> $= \sum_{1}^n \text{downtime of all standalone PaaS services during the measurement interval}$ [For example, if there are 20 PaaS services, the downtimes recorded automatically by the monitoring script as per the process defined in "Measurement for standalone PaaS service" above will be summed up (i.e., cumulative downtime will be computed)] Planned Downtime for standalone PaaS services = 3 hours per quarter Total Time = 24 X 7 measured over a period of one calendar month. Formula - 4: Availability % for containers = {1- [(Total Downtime-Planned downtime) / (Total Time – Planned Downtime)]} *100 <i>Total cumulative downtime</i> $= \sum_{1}^n \text{downtime of all standalone containers during the measurement interval}$ [For example, if there are 2,000 standalone containers, the downtimes recorded automatically by the monitoring script as per the process defined in "Measurement for standalone containers" above will be summed up (i.e., cumulative downtime will be computed)] Planned Downtime for standalone containers = 1 hours per quarter Total Time = 24 X 7 measured over a period of one calendar month. Overall Availability of Physical Server or Virtual Machine, Container or PaaS Service = ROUND ([Availability % for Physical Machines + Availability % for

SLA Description	Availability of Physical Server or Virtual Machine, Container or PaaS Service																								
	Virtual Machines + Availability % for PaaS services + Availability % for containers]/4,2) In the computation of the above formula, the Microsoft Excel ROUND () function or its programmatic functional equivalent will be used.																								
SLA target and Severity level for computation of Liquidated Damages	<table> <tr> <th>SLA Level</th><th>Severity Level</th></tr> <tr> <td>>=99.95%</td><td>None</td></tr> <tr> <td><99.95% and >= 99.5%</td><td>1</td></tr> <tr> <td><99.5% and >= 99%</td><td>2</td></tr> <tr> <td><99% and >=98%</td><td>3</td></tr> <tr> <td><98% and >=97%</td><td>4</td></tr> <tr> <td><97% and >=96%</td><td>5</td></tr> <tr> <td><96% and >=95%</td><td>6</td></tr> <tr> <td><95% and >=90%</td><td>7</td></tr> <tr> <td><90% and >=85%</td><td>8</td></tr> <tr> <td><85% and >=80%</td><td>9</td></tr> <tr> <td><80%</td><td>10</td></tr> </table>	SLA Level	Severity Level	>=99.95%	None	<99.95% and >= 99.5%	1	<99.5% and >= 99%	2	<99% and >=98%	3	<98% and >=97%	4	<97% and >=96%	5	<96% and >=95%	6	<95% and >=90%	7	<90% and >=85%	8	<85% and >=80%	9	<80%	10
SLA Level	Severity Level																								
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<80%	10																								
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in open source EMS tools.																								
Measurement interval	Monthly																								
Reporting interval	Quarterly																								
Reports or Data to be submitted, if any	Automated SLA dashboard in EMS tool																								

1.5.4.2 To-Be SLA 02 - Availability of Clustered Physical Server or Virtual Machine, Container or Service

SLA Description	Availability of Clustered Physical Server or Virtual Machine, Container or HA PaaS Service
Scope of SLA	Applies to clustered physical server, virtual machine, container or HA PaaS service
Tools, methods or scripts to be used to measure SLA	EMS tools to be implemented by MSIP – a custom monitoring script shall be executed once every minute from the SLA monitoring server.
Process to capture data or measure the parameter	In order to measure availability, it is not just sufficient to have the system is up and running. The MSIP can either use an agentless mechanism using SSH from the remote management servers or use an agent based solution for measuring the SLAs. Measurement for Clustered Physical Server or VM or Container

SLA Description	Availability of Clustered Physical Server or Virtual Machine, Container or HA PaaS Service																								
	Certain services/applications can be configured to operate in a HA cluster or in a group of nodes which are load balanced. In this situation, the following measurement techniques apply: - Availability of each node in the group is measured as defined in the Standalone Node Availability - The node level availability check is applied at the cluster level IP/URL and it has to meet the cluster level availability SLA Measurement for HA PaaS service - From the SLA monitoring server(s) connect via JDBC/HTTP(s)/TCP and invoke the service under measurement with load balancer IP or cluster IP - The service should return successfully with the response time within 120% of the baselined response time - The response time measurements will be applicable for 99 percentiles of the requests The monitoring script shall be executed once every minute from the SLA monitoring server.																								
SLA calculation	Formula: Availability % = {1- [(Total Downtime-Planned downtime) / (Total Time – Planned Downtime)]} *100 Planned Downtime -Total planned downtime as defined and agreed upon by MSIP and UIDAI. Total Time - 24 X 7 measured over a period of one calendar month.																								
SLA target and Severity level for computation of Liquidated Damages	<table> <tr> <th>SLA Level</th><th>Severity Level</th></tr> <tr> <td>>=99.99%</td><td>None</td></tr> <tr> <td><99.99% and >= 99.5%</td><td>1</td></tr> <tr> <td><99.5% and >= 99%</td><td>2</td></tr> <tr> <td><99% and >=98%</td><td>3</td></tr> <tr> <td><98% and >=97%</td><td>4</td></tr> <tr> <td><97% and >=96%</td><td>5</td></tr> <tr> <td><96% and >=95%</td><td>6</td></tr> <tr> <td><95% and >=90%</td><td>7</td></tr> <tr> <td><90% and >=85%</td><td>8</td></tr> <tr> <td><85% and >=80%</td><td>9</td></tr> <tr> <td><80%</td><td>10</td></tr> </table>	SLA Level	Severity Level	>=99.99%	None	<99.99% and >= 99.5%	1	<99.5% and >= 99%	2	<99% and >=98%	3	<98% and >=97%	4	<97% and >=96%	5	<96% and >=95%	6	<95% and >=90%	7	<90% and >=85%	8	<85% and >=80%	9	<80%	10
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<95% and >=90%	7																								
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Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.																								
Measurement interval	Monthly																								
Reporting interval	Quarterly																								

SLA Description	Availability of Clustered Physical Server or Virtual Machine, Container or HA PaaS Service
Reports or Data to be submitted, if any	Automated SLA dashboard in EMS tool

1.5.4.3 To-Be SLA 03 - Availability of EMS solution and SLA monitoring tool and dashboard

SLA Description	Availability of EMS solution and SLA monitoring tool and dashboard		
Scope of SLA	Applies to the EMS solution to be implemented by MSIP		
Tools, methods or scripts to be used to measure SLA	Custom script to be implemented by MSIP along with EMS tool		
Process to capture data or measure the parameter	<u>Measurement for EMS solution</u> The EMS Solution shall be configured to operate in a HA cluster. In this situation, the following measurement techniques apply: - Availability of the EMS service is measured from a Management server which invokes a synthetic EMS transaction on the EMS Dashboard (the synthetic transaction is created by record and replay of typical EMS transactions – like logging into the EMS Portal and checking the status of the various services on the dashboard) - The synthetic transaction is invoked once every minute. The EMS service is deemed to have a down time if either the script times out or returns a failure. The down time is measured from the last failed script to the next successful script execution. <u>SLA monitoring tool and dashboard</u> A custom script shall be developed which will create synthetic transactions simulating key SLA Dashboard operations/workflows. The script shall run from a remote server. The downtime is measured from the last unsuccessful script execution time to the next successful script execution time (a successful script execution is when there are no errors in all the steps of the process and there are no timeouts and all operations are completed within the designed response time).		
SLA calculation	Will be calculated for EMS solution and SLA monitoring tool separately using the below formula. Formula: Availability % = {1- [(Total Downtime-Planned downtime) / (Total Time – Planned Downtime)]} *100 Planned Downtime -Total planned downtime as defined and agreed upon by MSIP and UIDAI. Total Time - 24 X 7 measured over a period of one calendar month.		
SLA target	<table border="1"> <tr> <td>SLA Level</td><td>Severity Level</td></tr> </table>	SLA Level	Severity Level
SLA Level	Severity Level		

SLA Description	Availability of EMS solution and SLA monitoring tool and dashboard	
	>=99.99%	None
	<99.99% and >= 99.5%	1
	<99.5% and >= 99%	2
	<99% and >=98%	3
	<98% and >=97%	4
	<97% and >=96%	5
	<96% and >=95%	6
	<95% and >=90%	7
	<90% and >=85%	8
	<85% and >=80%	9
	<80%	10
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.	
Measurement interval	Monthly	
Reporting interval	Quarterly	
Reports or Data to be submitted, if any	Automated SLA dashboard in EMS tool. Availability graphs with the results of every script execution showing the status of service, timeouts, response times. This graph shall be provided for all SLA Parameters in the dashboard along with the downtime and availability calculations for all critical dashboard operations. This report shall be available online for real time analysis (for any selected date and time range) along with weekly, monthly, canned reports in xls and PDF formats.	

1.5.4.4 To-Be SLA 04 - Availability of CRM Solution and Contact Center Infrastructure

SLA Description	Availability of CRM and Contact Center infrastructure
Scope of SLA	Applies to the CRM Solution and Contact Center Infrastructure to be implemented by MSIP Contact Centre Infrastructure will cover the IVRS / VRU, the ACD, the CTI, the PBX, trunk lines, CRM software solution and VOIP gateway along with all associated server, storage, network and security components and peripherals.
Tools, methods or scripts to be used to measure SLA	Custom script to be implemented by MSIP
Process to capture data or measure the parameter	Measurement for CRM solution CRM Solution shall be configured to operate in a HA cluster. In this situation, the following measurement techniques apply: Availability of the CRM service is measured from a Management server which invokes a synthetic CRM transaction on the service. The synthetic transaction involves logging in to the CRM as a user and doing CRM transactions – logging a request, checking the status of a request.



SLA Description	Availability of CRM and Contact Center infrastructure																								
	ii. The synthetic transaction is invoked once every minute. The service is deemed to have a down time if either the script times out (time out should be less than or equal to 120% of the baselined response time) or returns a failure. The down time is measured from the last failed script to the next successful script execution. Measurement for Contact Center (“CC”) Infrastructure Contact Center Infrastructure shall be configured to operate in a HA cluster. In this situation, the following measurement techniques apply: i. Availability of the CC infrastructure is measured from a Management server which invokes a synthetic transaction on the service. It can be a simulated call to the Toll-Free Number (TFN), calls to be handled by the IVRS system, simulated messages to the chatbot organized such that end to end availability of the CC infrastructure is assessed. ii. The synthetic transaction is invoked once every minute. The service is deemed to have a down time if either the script times out (time out should be less than or equal to 120% of the baselined response time) or returns a failure. The down time is measured from the last failed script to the next successful script execution.																								
SLA calculation	Formula: Availability % = {1- [(Total Downtime-Planned downtime) / (Total Time – Planned Downtime)]} *100 Planned Downtime -Total planned downtime as defined and agreed upon by MSIP and UIDAI. Total Time - 24 X 7 measured over a period of one calendar month.																								
SLA target and Severity level for computation of Liquidated Damages	<table border="1"> <thead> <tr> <th>SLA Level</th><th>Severity Level</th></tr> </thead> <tbody> <tr> <td>>=99.99%</td><td>None</td></tr> <tr> <td><99.99% and >= 99.5%</td><td>1</td></tr> <tr> <td><99.5% and >= 99%</td><td>2</td></tr> <tr> <td><99% and >=98%</td><td>3</td></tr> <tr> <td><98% and >=97%</td><td>4</td></tr> <tr> <td><97% and >=96%</td><td>5</td></tr> <tr> <td><96% and >=95%</td><td>6</td></tr> <tr> <td><95% and >=90%</td><td>7</td></tr> <tr> <td><90% and >=85%</td><td>8</td></tr> <tr> <td><85% and >=80%</td><td>9</td></tr> <tr> <td><80%</td><td>10</td></tr> </tbody> </table>	SLA Level	Severity Level	>=99.99%	None	<99.99% and >= 99.5%	1	<99.5% and >= 99%	2	<99% and >=98%	3	<98% and >=97%	4	<97% and >=96%	5	<96% and >=95%	6	<95% and >=90%	7	<90% and >=85%	8	<85% and >=80%	9	<80%	10
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<90% and >=85%	8																								
<85% and >=80%	9																								
<80%	10																								
	3																								
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.																								

SLA Description	Availability of CRM and Contact Center infrastructure
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Automated SLA dashboard in EMS tool. Availability graphs with the results of every script execution showing the status of service, timeouts, response times. This graph shall be provided for all SLA Parameters in the dashboard along with the downtime and availability calculations for all critical dashboard operations. This report shall be available online for real time analysis (for any selected date and time range) along with weekly, monthly, canned reports in xls and PDF formats.

1.5.4.5 To-Be SLA 05 - Availability of Email solution

SLA Description	Availability of Email																								
Scope of SLA	Applies to the Email to be implemented by MSIP																								
Tools, methods or scripts to be used to measure SLA	Custom script to be implemented by MSIP																								
Process to capture data or measure the parameter	Measured every 5 minutes by connecting to the e-mail gateway (by a synthetic client) by logging into the e-mail server. The email solution is down if the synthetic client is either not able to connect or the login fails. The down time is measured from the last failed script to the next successful script execution.																								
SLA calculation	Formula: Availability % = {1- [(Total Downtime-Planned downtime) / (Total Time – Planned Downtime)]} *100 Planned Downtime -Total planned downtime as defined and agreed upon by MSIP and UIDAI. Total Time - 24 X 7 measured over a period of one calendar month.																								
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<95% and >=90%	7																								
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<80%	10																								

SLA Description	Availability of Email
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Automated SLA dashboard in EMS tool. Availability graphs with the results of every script execution showing the status of service, timeouts, response times. This graph shall be provided for all SLA Parameters in the dashboard along with the downtime and availability calculations for all critical dashboard operations. This report shall be available online for real time analysis (for any selected date and time range) along with weekly, monthly, canned reports in xls and PDF formats.

1.5.4.6 To-Be SLA 06 - Availability of SMS Gateway

SLA Description	Availability of SMS Gateway																		
Scope of SLA	Applies to the SMS Gateway to be implemented by MSIP																		
Tools, methods or scripts to be used to measure SLA	Custom script to be implemented by MSIP																		
Process to capture data or measure the parameter	Measured every 5 minutes by connecting to the SMS gateway (by a synthetic client) by invoking the rest API and sending a synthetic SMS message, The SMS gateway is down if the client is either not able to connect or send an SMS successfully. The down time is measured from the last failed script to the next successful script execution.																		
SLA calculation	Formula: Availability % = {1- [(Total Downtime-Planned downtime) / (Total Time – Planned Downtime)]} *100 Planned Downtime -Total planned downtime as defined and agreed upon by MSIP and UIDAI. Total Time - 24 X 7 measured over a period of one calendar month.																		
SLA target and Severity level for computation of Liquidated Damages	<table> <tr> <th>SLA Level</th><th>Severity Level</th></tr> <tr> <td>>=99.95%</td><td>None</td></tr> <tr> <td><99.95% and >= 99.5%</td><td>1</td></tr> <tr> <td><99.5% and >= 99%</td><td>2</td></tr> <tr> <td><99% and >=98%</td><td>3</td></tr> <tr> <td><98% and >=97%</td><td>4</td></tr> <tr> <td><97% and >=96%</td><td>5</td></tr> <tr> <td><96% and >=95%</td><td>6</td></tr> <tr> <td><95% and >=90%</td><td>7</td></tr> </table>	SLA Level	Severity Level	>=99.95%	None	<99.95% and >= 99.5%	1	<99.5% and >= 99%	2	<99% and >=98%	3	<98% and >=97%	4	<97% and >=96%	5	<96% and >=95%	6	<95% and >=90%	7
SLA Level	Severity Level																		
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SLA Description	Availability of SMS Gateway	
	<90% and >=85%	8
	<85% and >=80%	9
	<80%	10
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.	
Measurement interval	Monthly	
Reporting interval	Quarterly	
Reports or Data to be submitted, if any	Automated SLA dashboard in EMS tool. Availability graphs with the results of every script execution showing the status of service, timeouts, response times. This graph shall be provided for all SLA Parameters in the dashboard along with the downtime and availability calculations for all critical dashboard operations. This report shall be available online for real time analysis (for any selected date and time range) along with weekly, monthly, canned reports in xls and PDF formats.	

1.5.5 To-Be SLA 07 - Provisioning SLAs

SLA Description	Provisioning SLAs
Scope of SLA	Covers physical servers, virtual machines and containers
Tools, methods or scripts to be used to measure SLA	The provisioning SLA metrics are used to measure the time taken to provision a service after due approvals either via the EMS portal, command line scripts or APIs. The detailed steps/workflow to be used in provisioning are defined below. - The MSIP shall implement an automated workflow for provisioning the IaaS and PaaS services as requested by the users. The provisioning request may originate as a request via the EMS portal or via a script or API invoked by application services or command line tools. In either case, after the due approval process, which may involve multi-level approvals, the provisioning workflow shall be executed. - The MSIP shall co-ordinate with MSAP, BSP, UIDAI and other service users to define the template for the provisioning workflows. - Once a service is provisioned, it shall be fully operational and ready for production use and shall be subjected to the availability, throughput and response time SLAs. - Below is a typical provisioning workflow for provisioning a PaaS Service:



SLA Description	Provisioning SLAs						
	a. Provision one or more Physical, Virtual machines, Containers with required CPUs/Cores, Memory and Storage, OS as per the defined provisioning template b. Harden the system as per UIDAI defined security policies and procedures c. Apply relevant patches or updates d. Install and configure all agents – including backup, antivirus, logging, monitoring, management e. If any service is agentless, enable appropriate ports in the firewall for remote access for the management system f. Configure IP addresses g. Enable requisite ports in firewall for all inbound and outbound communications h. Configure the load balancers if the node is in a load balanced cluster i. Tune all system parameters like max connections, timeouts, maxfiles, processes, threads, shared memory, semaphores or any other parameter as required by the service j. Enable appropriate file access permissions and privileges required for the applications k. Install all relevant binaries, config files, scripts for the PaaS Service l. Run a system health check to ensure that the system has sufficient resources to is functioning correctly. m. Run and execute PaaS service health and sanity tests to ensure that the service is up and running. n. Check the connectivity to all the external dependent hosts, services, URLs						
Process to capture data or measure the parameter	a. The provisioning workflow process will log a timestamp at the beginning of the process and again at the end of the process. b. The difference between the provisioning process end time stamp and the start time stamp will be the time taken for the provisioning. c. The provisioning process time measurements will start after all approvals (if any) are completed.						
SLA calculation	<u>For calculation of SLA, simple average of measurements at 99 percentile of provisioning results for the given measurement period will be considered.</u> This will be computed separately for VM, containers and physical servers and simple average of the three will be considered.						
SLA target and Severity level for computation of Liquidated Damages	<table> <tr> <th>SLA Measured for VM (in minutes)</th><th>Severity Level</th></tr> <tr> <td><=5 minutes</td><td>None</td></tr> <tr> <td>>5 minutes and <=6 minutes</td><td>1</td></tr> </table>	SLA Measured for VM (in minutes)	Severity Level	<=5 minutes	None	>5 minutes and <=6 minutes	1
SLA Measured for VM (in minutes)	Severity Level						
<=5 minutes	None						
>5 minutes and <=6 minutes	1						

SLA Description	Provisioning SLAs			
	>6 minutes and <=7 minutes		2	
	>7 minutes and <=8 minutes		3	
	>8 minutes and <=9 minutes		4	
	>9 minutes and <=10 minutes		5	
	>10 minutes and <=15 minutes		6	
	>15 minutes		7	
	SLA Measured for Containers (in seconds)		Severity Level	
	<=30 seconds		None	
	>30 seconds and <=45 seconds		1	
	>45 seconds and <=60 seconds		2	
	>60 seconds and <=75 seconds		3	
	>75 seconds and <=90 seconds		4	
	>90 seconds and <=120 seconds		5	
	>120 seconds and <=150 seconds		6	
	>150 seconds and <=180 seconds		7	
	>180 seconds and <=210 seconds		8	
	>210 seconds and <=240 seconds		9	
	>240 seconds		10	
	SLA Measured for Physical Servers (in hours) #		Severity Level	
	<=6 hours		None	
	>6 hours and <=7 hours		1	
	>7 hours and <=8 hours		2	
	>8 hours and <=9 hours		3	
	>9 hours and <=10 hours		4	
	>10 hours and <=11 hours		5	
	>11 hours and <=12 hours		6	
	>12 hours and <=13 hours		7	
	>13 hours and <=14 hours		8	
	>14 hours and <=15 hours		9	
	>15 hours		10	
	Note: # - Provisioning SLA for physical servers is subject to availability of unutilized physical servers in the MSIP's inventory. The start time for computation of the SLA will be from the time of approval of the provisioning request by UIDAI after confirmation of availability of physical servers in inventory.			
	Overall provisioning SLA related LD for a measurement interval = Simple Average of LD computed for VMs, Containers and Physical Servers			
	Example:			
	Type	Simple average of measurements at 99 percentile of provisioning results for the	Severity Level	% of LD applicable

SLA Description	Provisioning SLAs			
		given measurement period		
	Virtual Machine	7 minutes 24 seconds	3	0.3%
	Container	43 seconds	1	0.1%
	Physical Machine	8 hours 24 minutes	3	0.3%
	LD applicable for the measurement interval			=ROUND ([0.3% + 0.1% + 0.3%]/3,2) = 0.02%
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.			
Measurement interval	Monthly			
Reporting interval	Quarterly			
Reports or Data to be submitted, if any	Automated SLA dashboard in EMS tool			

1.5.6 To-Be SLA 08 - Performance SLAs

SLA Description	Performance SLAs
Scope of SLA	The performance SLA's are used to measure the throughput, response time of all IaaS and PaaS Services
Tools, methods or scripts to be used to measure SLA	<u>VM Performance</u> The VM performance is measured by executing the benchmark script. The benchmark script is executed in the benchmarking environment and the benchmark results are baselined for the configured VM. <u>PaaS services</u> The PaaS Service performance is measured by executing the PaaS Service Benchmark script. The benchmark script is executed in the benchmarking environment and the benchmark results are baselined for the configured infrastructure.

SLA Description	Performance SLAs
Process to capture data or measure the parameter	<u>VM Performance</u> The tests will be executed on all the VMs. The benchmarking tests will compute the CPU/memory performance and network/IO performance. <u>PaaS performance</u> The PaaS service throughput and response times shall be measured from the log files of the corresponding service.
SLA calculation	The benchmark script is executed automatically at least once in 30 days in production environment and compared with the baseline results. MSIP will have to prepare the benchmarking scripts as per UIDAI's guidelines.
SLA target	For VMs - Any reduction in benchmark results by more than 20% is a violation of the SLA. The SLA measurements shall cover the 99 percentiles of the benchmarking results for the given measurement period. For PaaS services - Any reduction in throughput (proportionally for the given load) and/or response time results by more than 20% is a violation of the SLA. The SLA measurements shall cover the 99 percentiles of the benchmarking results for the given measurement period.
Severity level for computation of Liquidated Damages	2
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Automated SLA dashboard in EMS tool

1.5.7 Private Cloud Storage SLAs

1.5.7.1 To-Be SLA 09 - Compliance to backup schedule

The backup and recovery SLAs are defined to measure the compliance and performance of the backup and recovery processes. The SLA dashboard shall show the status of these SLAs in real-time.

SLA Description	Private Cloud Storage - Backup and restore
Scope of SLA	Private cloud storage
Tools, methods or scripts to be used to measure SLA	Custom scripts integrated with SLA Management Tool

SLA Description	Private Cloud Storage - Backup and restore
Process to capture data or measure the parameter	Logs from the backup tool showing the date, time stamp (start and end time), device label/id, content backed up, size of backup
SLA calculation	NA
SLA target	100%
Severity level for computation of Liquidated Damages	3
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Monthly report of the analysis of the backup logs. This report shall be available online for real time analysis (for any selected date and time range) along with weekly, monthly, canned reports in xls and PDF formats.

1.5.7.2 To-Be SLA 10 - Backup throughput

SLA Description	Private Cloud Storage - Backup throughput
Scope of SLA	Private cloud storage
Tools, methods or scripts to be used to measure SLA	Custom scripts integrated with SLA Management Tool
Process to capture data or measure the parameter	Logs from the backup tool showing the date, time stamp (start and end time), device label/id, content backed up, size of backup
SLA calculation	Average of daily backup throughput measured over a period of 1 month To be computed through custom scripts implemented by MSIP
SLA target	300 TB per 8 hours
Severity level for computation of Liquidated Damages	2
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Audit Report showing the device label, location. This report shall be available online for real time analysis (for any selected date and time range) along with weekly, monthly, canned reports in xls and PDF formats.

SLA Description	Private Cloud Storage - Backup throughput

1.5.7.3 To-Be SLA 11 - Time taken for daily backups

SLA Description	Private Cloud Storage – Time taken for daily backups
Scope of SLA	Private cloud storage
Tools, methods or scripts to be used to measure SLA	Custom scripts integrated with SLA Management Tool
Process to capture data or measure the parameter	Logs from the backup tool showing the date, time stamp (start and end time), device label/id, content backed up, size of backup
SLA calculation	Average of daily backup time taken measured over a period of 1 month To be computed through custom scripts implemented by MSIP
SLA target	8 hours for successful completion of daily backup
Severity level for computation of Liquidated Damages	2
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Automated SLA dashboard in EMS tool

1.5.7.4 To-Be SLA 12 - Storage response time

SLA Description	Private Cloud Storage – Storage response time
Scope of SLA	Private cloud storage
Tools, methods or scripts to be used to measure SLA	Custom scripts integrated with SLA Management Tool
Process to capture data or measure the parameter	A custom script shall be provided by MSIP using the API/SDK provided by the storage array/solution and shall be run on each storage array/host for the I/O requests.
SLA calculation	A custom script shall also be provided by MSIP which will run on each host to measure the i/o latency, service time - by using iostat or equivalent command. The script shall be run once every 5 minutes.

SLA Description	Private Cloud Storage – Storage response time
	Percentile shall be calculated on all measurements taken during a measurement interval. Values greater than the 99.9 percentile level will be ignored in the calculation.
SLA target	99.9 percentile < 5 ms for SSD Storage 99.9 Percentile < 20 ms for SAS/SATA Storage
Severity level for computation of Liquidated Damages	3
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Monthly report of IO response time by Storage Server, Volume and by host/vm with distribution graph of I/O response times shall be provided. This report shall be available online for real time analysis (for any selected date and time range) along with weekly, monthly, canned reports in xls and PDF formats.

1.5.7.5 To-Be SLA 13 - Storage throughput

SLA Description	Private Cloud Storage – Storage throughput - Disks (in IOPS)
Scope of SLA	Private cloud storage
Tools, methods or scripts to be used to measure SLA	Custom scripts integrated with SLA Management Tool
Process to capture data or measure the parameter	A custom script shall be provided by MSIP using the API/SDK provided by the storage array/solution and shall be run on each storage array/server for the I/O throughput. A custom script shall also be provided which will run on each host to measure the IOPS - by using iostat or equivalent command An IO baseline shall be created before a new storage array/server and volume/lun is provisioned and further throughput shall be compared with the baseline, the baseline throughput shall be inline with the predefined/configured IOPS for the specific storage server/volume. The Baseline value will be determined when the storage volume is provisioned by using tools like: i) iobench - io benchmarking open source tool OR ii) scripts (using dd command to read write data of different sizes 50 MB, 1 GB)) and the iostat command is used to measure the iops of the volume/disk while the dd script is running.

SLA Description	Private Cloud Storage – Storage throughput - Disks (in IOPS)
SLA calculation	NA
SLA target	IOPS as per defined configuration/specification with not more than +/- 20% deviation from the baseline.
Severity level for computation of Liquidated Damages	2
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Monthly report of IO throughput Storage Server, Volume and by host/vm along with distribution graph of IOPS (read and writes) and deviation from baseline shall be provided. This report shall be available online for real time analysis (for any selected date and time range) along with weekly, monthly, canned reports in xls and PDF formats.

1.5.7.6 To-Be SLA 14 - Storage data loss

SLA Description	Private Cloud Storage – Storage data loss
Scope of SLA	Private cloud storage. It is very important for the MSIP to ensure that the tools, processes, monitoring is in place to proactively identify issues which can cause data loss. Data loss can be measured at the host level and at the storage device level (SAN, Software Defined Storage, NAS).
Tools, methods or scripts to be used to measure SLA	- A cron/scheduled job shall be run on every OS instance at least once in 24 hours. - A cron/scheduled job shall be run on every storage device least once in 24 hours.
Process to capture data or measure the parameter	<u>Host level data loss</u> The host level data loss is measured by checking for bad blocks in the system logs. A cron/scheduled job shall be run on every OS instance at least once in 24 hours. The system logs of all hosts have to be scanned at least once a day to check for any block errors, bad blocks. Bad blocks or Block errors in any host partition block which contain data of data bases, data stores, object stores, file stores, EMS Applications, PaaS service data for which there is no error free copy is treated as a data loss.

SLA Description	Private Cloud Storage – Storage data loss
	The MSIP should ensure that all data is backed up frequently and store multiple copies to prevent data loss. The SLA dashboard should display host wise, day wise, month wise bad blocks. <u>Storage level data loss</u> The storage loss is measured by checking for bad blocks in the storage subsystem (SAN, NAS or Software Defined Storage, Archival Storage) A cron/scheduled job shall be run on every storage device least once in 24 hours. Storage device analysis tools report the disk failures, bad blocks and any other storage/media errors. The storage analysis tools should be run once every day and the report should be available on the dashboard – device wise, host wise, date wise. The MSIP shall proactively identify such issues and take corrective action to prevent any data loss. The data loss at the storage device level may lead to host level bad blocks or data errors.
SLA calculation	NA
SLA target	Any non-zero data loss is a violation of the SLA.
Severity level for computation of Liquidated Damages	5
Assumptions, if any	The MSIP shall take appropriate action to prevent data loss by monitoring the bad blocks in the storage device.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	The SLA dashboard shall provide day wise, host wise, device wise aggregate bad blocks count with drill down.

1.5.7.7 To-Be SLA 15 - Replication Lags

SLA Description	Replication lag
Scope of SLA	This SLA covers DC to DC link This SLA is provided to ensure that the MSIP provided network and associated infrastructure does not contribute to the lag of the replication.
Tools, methods or scripts to be used to measure SLA	SLA monitoring tool - The replication lag is measured by the APIs or Command available from each data base or data store.

SLA Description	Replication lag
Process to capture data or measure the parameter	All data has to be replicated within each DC and across DC to eliminate data loss in case of any disasters. The SLA monitoring tool should continuously monitor the replication lag across the data stores to ensure that it does not violate the SLA. The replication lag is measured by the APIs/Command available from each data base or data store.
SLA calculation	Formula: Simple average of all the replication lag times recorded over the measurement interval.
SLA targets	Replication Lag across Data Store copies within the same DC < 5 seconds Replication Lag across Data Stores across DC and DR sites < 30 seconds
Severity level for computation of Liquidated Damages	2
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in open source EMS tools.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Automated reporting in SLA dashboard. The SLA monitoring dashboard shall display the replication lag for each data store in real time and store all the historical data.

1.5.8 Network SLAs applicable for TSPs in To-Be state

1.5.8.1 To-Be SLA 16 - Network Uptime, Utilization and Performance

SLA Description	Network Uptime, Utilization and Performance
Scope of SLA	This SLA covers MPLS Network, Internet Links and Replication Network Note: The "Replication Network uptime" refers to the DC to DC network links uptime.
Tools, methods or scripts to be used to measure SLA	SNMP Client
Process to capture data or measure the parameter	All the network devices should provide SNMP V1/V2/V3 support. The network SLAs can be monitored by checking the status of the corresponding ports in all the devices. The SLA management tools should poll all the LAN

SLA Description	Network Uptime, Utilization and Performance
	and WAN devices to check for the link status (uptime), utilization and the packet losses once every 60 seconds.
SLA calculation	Downtime will be calculated separately for MPLS network uptime, Internet individual link uptime, P2P individual link uptime, MPLS individual link uptime, Internet uptime, Replication network uptime using the below formula. $\text{Availability \%} = \{1 - [(\text{Total Downtime} - \text{Planned downtime}) / (\text{Total Time} - \text{Planned Downtime})]\} * 100$ Planned Downtime – 1 hour per quarter Total Time - 24 X 7 measured over a period of one calendar month. For Network Packet Loss and WAN utilization, a simple average of all the data points recorded by the SLA management tools over the measurement interval will be taken and compared against the SLA targets.
SLA targets	The below targets are applicable to each of the individual Telecom Service Providers (TSP). Internet individual Link Uptime $\geq 98.00\%$ AND P2P individual Link Uptime $\geq 98.00\%$ AND MPLS individual Link Uptime $\geq 98.00\%$ AND Network packet loss $\leq 0.1\%$ Even if one of the above parameters are not met, the SLA would be considered to be breached and the LD will become applicable as per the severity level.
Severity level for computation of Liquidated Damages	2
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in open EMS tools.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Automated reporting in SLA dashboard

1.5.8.2 To-Be SLA 17 - Network Latency

SLA Description	Network Latency
Scope of SLA	This SLA covers MPLS Network, Internet Links and Replication Network

SLA Description	Network Latency
Tools, methods or scripts to be used to measure SLA	SNMP Client
Process to capture data or measure the parameter	The network latency is measured by a ping from the SLA management server to at least two or more LAN ports across each VLAN/subnets. The network utilization is measured by the SNMP variable port utilization of the corresponding device port. The network latency is WAN links is measured by a ping from the WAN Router in DC to WAN Router in DR and Vice Versa. This ping is repeated very 1 minute. The Network latency is the latency reported by each ping response time.
SLA calculation	Formula: Simple average of all the ping response times observed during the measurement interval for the WAN and LAN links respectively excluding times greater than the specified percentile level.
SLA targets	Network latency – LAN <5 milliseconds at 99.9 percentile. Network latency – WAN <60 milliseconds at 99.5 percentile
Severity level for computation of Liquidated Damages	2
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in open source EMS tools.
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Automated reporting in SLA dashboard

1.5.9 To-Be SLA 18 - P1 Service Desk (“SD”) ticket resolution time

Track / Category	Target	Applied Severity level
Service desk response time	>= 99%	None
	>= 98% and <99%	1
	>= 97% and <98%	2
	>= 95% and <97%	3
	>= 90% and <95%	4
	<90%	5
SLA Description	P1 Service Desk ticket resolution time	
Definition of SLA	Metric: % of Priority-1 tickets resolved within 30 mins	
Scope of SLA	The scope of this SLA includes response to P1 incidents raised in Service Desk	

Tool Used for SLA	Service Desk tool implemented by MSIP as part of EMS implementation
Data Points for SLA Calculation	Automated calculation by Service Desk tool implemented by MSIP as part of EMS implementation
SLA Calculation	Formula: [Number of P1 tickets resolved to within 30 min/ Total Number of P1 Tickets raised in that Month] *100 Note: Tickets with incomplete Service Description, Tickets erroneously logged by a user and subsequently cancelled by the user and Tickets erroneously auto-generated by a system will be excluded from the total number of Service Desk Tickets for calculation of these SLAs.
Measurement Interval	Monthly
Reporting Interval	Quarterly
Reports and data to be submitted	Automated Quarterly SLA Report in Service Desk tool implemented by MSIP as part of EMS implementation

1.5.10 To-Be SLA 19 - P2 Service Desk ticket resolution time

Track / Category	Target	Applied Severity level
Service desk response time	>= 99%	None
	>= 98% and <99%	1
	>= 97% and <98%	2
	>= 95% and <97%	3
	>= 90% and <95%	4
	<90%	5
SLA Description	P2 Service Desk ticket resolution time	
Definition of SLA	Metric: % of Priority-2 tickets resolved within 1 hour	
Scope of SLA	The scope of this SLA includes response to P2 incidents raised in Service Desk	
Tool Used for SLA	Service Desk tool implemented by MSIP as part of EMS implementation	
Data Points for SLA Calculation	Automated calculation in Service Desk tool implemented by MSIP as part of EMS implementation	
SLA Calculation	Formula: [Number of P2 tickets resolved within 1 hour/ Total Number of P2 Tickets raised in that Month] *100 Note: Tickets with incomplete Service Description, Tickets erroneously logged by a user and subsequently cancelled by the user and Tickets erroneously auto-	

	generated by a system will be excluded from the total number of Service Desk Tickets for calculation of these SLAs.
Measurement Interval	Monthly
Reporting Interval	Quarterly
Reports and data to be submitted	Automated Quarterly SLA Report in Service Desk tool implemented by MSIP as part of EMS implementation

1.5.11 To-Be SLA 20 - P3 Service Desk ticket resolved time

Track / Category	Target	Applied Severity level
Service desk response time	>=95%	0
	>=80% and <95%	1
	<80%	2
Definition of SLA	Metric: % of Priority-3 tickets resolved within 2 hours	
Scope of SLA	The scope of this SLA includes response to P3 incidents raised in Service Desk	
Tool Used for SLA	Service Desk tool implemented by MSIP as part of EMS implementation	
Data Points for SLA Calculation	Automated calculation in Service Desk tool implemented by MSIP as part of EMS implementation	
SLA Calculation	Formula: [Number of P3 tickets responded within 2 hours/ Total Number of P3 Tickets raised in that Month] *100 Note: Tickets with incomplete Service Description, Tickets erroneously logged by a user and subsequently cancelled by the user and Tickets erroneously auto-generated by a system will be excluded from the total number of Service Desk Tickets for calculation of these SLAs.	
Measurement Interval	Monthly	
Reporting Interval	Quarterly	
Reports and data to be submitted	Automated Quarterly SLA Report from Service Desk tool implemented by MSIP as part of EMS implementation	

1.5.12 To-Be SLA 21 - P4 Service Desk ticket resolution time

Track / Category	Target	Applied Severity level
Service desk response time	>=95%	0
	>=80% and <95%	1

	<80%	2
Definition of SLA	Metric: % of Priority-4 tickets resolved within 4 hours	
Scope of SLA	The scope of this SLA includes response to P4 incidents raised in Service Desk	
Tool Used for SLA	Service Desk tool implemented by MSIP as part of EMS implementation	
Data Points for SLA Calculation	Automated calculation in Service Desk tool implemented by MSIP as part of EMS implementation	
SLA Calculation	Formula: [Number of P4 tickets resolved within 4 hours/ Total Number of P4 Tickets raised in that Month] *100 Note: Tickets with incomplete Service Description, Tickets erroneously logged by a user and subsequently cancelled by the user and Tickets erroneously auto-generated by a system will be excluded from the total number of Service Desk Tickets for calculation of these SLAs.	
Measurement Interval	Monthly	
Reporting Interval	Quarterly	
Reports and data to be submitted	Quarterly SLA Report from Service Desk tool implemented by MSIP as part of EMS implementation	

1.5.13 To-Be SLA 22 - Turnaround time for changes to IVRS tree

SLA Description	IVRS tree changes
Scope of SLA	This SLA covers the IVRS system
Tools, methods or scripts to be used to measure SLA	Service Desk Tool
Process to capture data or measure the parameter	The elapsed time shall be computed as the interval between logging of IVRS Tree Change Requirement as a Ticket in the Service Desk Tool by UIDAI authorized representative for initiating changes to IVRS tree to the time of completion of the change and closure of the request in the Service Desk tool.
SLA calculation	Formula: Elapsed time = Time of Closure of Service Desk ticket for IVRS Tree Change – Time of Opening of Service Desk ticket for IVRS Tree Change Average elapsed time of all tickets opened during the month will be considered.
SLA targets	Elapsed Time <= 5 days

SLA Description	IVRS tree changes
Severity level for computation of Liquidated Damages	2
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Automated reporting in SLA dashboard.

1.5.14 To-Be SLA 23 - Planned Down Time SLAs

- A large system like CIDR will need down time to manage upgrades, patches or any other maintenance activities. The MSIP shall ensure that all such activities be carried out by notifying all stakeholders ahead of time and get the same approved by UIDAI.
- Following are the SLA targets and Severity levels for planned down times.

S. No	Parameter – Planned Down time	SLA target - Planned downtime duration not to exceed	Severity level
1	Authentication Infrastructure	0.5 Hours per quarter	5
2	Enrolment Infrastructure	3 Hours per quarter	5
3	CRM and CCM Infrastructure	6 Hours per quarter	5
4	Portal Infrastructure	1 Hours per quarter	5
5	All PaaS Services	3 Hours per quarter	5

- Note that in the above SLAs from 1 to 4, if any of the services use PaaS Services, the downtime SLA duration will be accounted only once. For example, the total planned down time for Enrolment Infrastructure shall be less than 3 Hours per quarter including the associated PaaS Services downtime.
- In case any of the above planned downtimes are not utilized during a given quarter, the MSIP may carry forward the downtimes. Such unutilized planned downtimes should be utilized within one year. For example, if planned downtimes are unutilized during Q1 of a financial year, such planned downtime can be carried forward until the end of Q4 of the same financial year. Similarly, if planned downtimes are not utilized during Q1 and Q3 of a financial year, the cumulative downtime available can be utilized until the end of Q4 of the same financial year. However, if any planned downtime is available in Q4 and not utilized in Q4, such downtime cannot be carried forward to the next financial year.
- On an exceptional basis, UIDAI upon written request from MSIP, provide additional planned downtime for a quarter over and above the SLA targets specified in (b) above to support any major planned changes.

1.5.15 To-Be SLA 24 - Traceability of Backup Tapes

Track / Category	Target	Applied Severity level
Backup & Restore – Traceability of Backup Devices	≥100%	0
	<100%	5

Definition of SLA	Metric: Exact location for any backup tape containing a specific data should be known in the real time
Scope of SLA	The scope includes all the backup tapes
Tool Used for SLA	EMS Tool implemented by MSIP
Data Points for SLA Calculation	Report for the location of any backup tape. Online report for the tapes currently in use, and within the backup tape library Offline report for the tapes which are currently not in the tape library
SLA Calculation	The location of backup tapes to be reported to UIDAI for each DC on monthly basis. The data submitted will be verified against actual location of the tapes and SLA will be calculated as per the below formula. $\% \text{ Compliance} = [\text{Number of backup tapes whose location is verified to be correct against data available in EMS tool} / \text{Total number of backup tapes whose location is verified against data available in EMS tool}]$
Assumptions	NA
Measurement Interval	Monthly
Reporting Interval	Quarterly
Reports and data to be submitted	Quarterly SLA Report A Summary sheet shall capture inventory of Tapes (used and spare) and following reports shall be submitted for used tapes. • Online Report • Fireproof safe report • Offline Report • Mapping of backup data with media tape

1.5.16 To-Be SLA 25 - Monthly testing and Validation of Backup Devices

Track / Category	Target	Applied Severity level
Backup & Restore – Monthly testing and Validation of Backup Devices	=100%	0
	<100%	3
Definition of SLA	Monthly testing and validation of backup devices for the quality of data	
Scope of SLA	The scope includes all the backup tapes	
Tool Used for SLA	EMS Tool and Backup and Restore Tool implemented by MSIP along with EMC Networker and Symantec NetBackup	

Data Points for SLA Calculation	Restoration logs from Backup and Restore Tool implemented by MSIP, EMC networker and Symantec NetBackup Mail confirmation from Application owner on the successful completion of restoration.
SLA Calculation	The restoration testing will be performed on sample basis as per the backup testing and restoration process. Any deviation or non-compliances observed in testing and validation of the backup devices from the testing and restoration process will result in the breach of the SLA. Compliance % = [Number of successful tests and validation on backup devices/Total number of devices in sample selected for tests and validation]
Assumptions	The sample size (5% of used storage for each period i.e. daily, weekly etc.) for restoration shall be taken as per backup testing and restoration process. Restoration shall be done in the restoration environment. Certain fixed size of data will be restored from each backup and then given to Application owner for testing Backup team shall ensure that the backed-up data has been restored without any errors. Data validation and integrity check of the restore is performed by the application owner and the confirmation to be provided to backup team.
Measurement Interval	Monthly
Reporting Interval	Quarterly
Reports and data to be submitted	Restoration testing report and confirmation from respective UIDAI's stakeholders for successful restoration testing

1.5.17 To-Be SLA 26 - Re-triggering of faulty backups

Track / Category	Target	Applied Severity level
Backup & Restore – Reporting of faulty backups	=100%	0
	<100%	4
Definition of SLA	The backups tested faulty should be re-backed up within 24 hours of testing	
Scope of SLA	The scope includes the faulty backup during the reporting period	
Tool Used for SLA	EMS Tool implemented by MSIP, Backup tool implemented by MSIP along with existing EMC Networker and Symantec Netbackup tools.	
Data Points for SLA Calculation	Failed backups are re-triggered within 24 hours of failure. Logs are provided to confirm the retriggering of the backup along with the completion timestamp.	



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SLA Calculation	Formula for SLA calculation = Total no. of retriggered faulty backups reported in the last 24 hours/ No. of faulty backups in the last 24 hours For a given month, the average of all daily observations will be taken for computation of this SLA.
Measurement Interval	Monthly
Reporting Interval	Quarterly
Reports and data to be submitted	Quarterly SLA Report

1.5.18 To-Be SLA 27 - Inventory Management - Asset Change or Move Reporting

Track / Category	Target	Applied Severity level
Inventory Management	= 100%	None
	< 100%	4
SLA Description	Asset Change or Move Reporting	
Definition of SLA	Metric: Any asset change or move should be reported within 24 hours in the prescribed format and updated in ITAM module of EMS tool implemented by MSIP	
Scope of SLA	The scope includes any asset change or move during the reporting period	
Tool Used for SLA	IT Asset Management ("ITAM") module of EMS tool implemented by MSIP	
Data Points for SLA Calculation	Install Move, Add, and Change (IMAC) is tracked in Service desk and the data for SLA computation, i.e., IMAC resolution and IMAC activity completion is also captured from tool. All IMAC requests are pre-approved by UIDAI stakeholder in IMAC tool before the change gets implemented. Once the activity is completed, the corresponding IMAC ticket is closed and automated email notification is sent to UIDAI. UIDAI is informed both before and after asset change/move. Asset Change will also include any changes in status of AMC, expiry of warranty coverage, any changes in configuration of the system. UIDAI reserves the right to audit the quality of data update done in ITAM and any errors detected in data updated in ITAM will be considered as breach of SLA. This will include but not be limited to the following fields: 1) Area 2) Rack Number 3) Row Number 4) Enclosure or Chassis Serial Number 5) Asset Name 6) Manufacturer Name 7) Model Name 8) Serial Number 9) Asset Category 10) Asset Type 11) Location Name 12) Status 13) Asset Lifecycle Status 14) Owner 15) Host Name 16) IP Address 17) Asset Custodian 18) Purchase Order ID 19) Purchase Order Date 20) Hard Disk Drive ("HDD") Count 21) HDD Capacity 22) RAM Capacity 23) Processor Count	

	24) Processor Type 25) Operating System 26) Warranty Start Date 27) Warranty End Date 28) Invoice Number 29) Invoice Date 30) Date of arrival in store 31) Date when connected to operations 32) Asset Criticality, CIA rating 33) Risk assessment status 34) Retention period 35) Application hosted 36) Application Name 37) Physical or Virtual Asset 38) Last Updated Date
SLA Calculation	SLA calculation formula= Total IMAC Tickets closed during the period and for which ITAM is updated across all DCs during the period/ Total approved IMAC Tickets raised across DCs during the period] *100
Measurement Interval	Monthly
Reporting Interval	Quarterly
Reports and data to be submitted	Quarterly SLA Report

1.5.19 To-Be SLA 28 - Location of asset in real time and Asset Traceability

Track / Category	Target	Applied Severity level
Inventory Management	=100%	None
	<100%	4
SLA Description	Location of asset in real time and Asset Traceability	
Definition of SLA	Metric: Exact location for any asset listed in asset database (ITAM) should be known in the real time. How: Random Tracking Test will be carried out	
Scope of SLA	The scope includes all the assets in Manesar and Bengaluru Data centres	
Data Points for SLA Calculation	A monthly review will be conducted to verify the location of the assets as per ITAM database. The review will be conducted on a sample basis. Physical Verification (Random Testing) Procedure: UIDAI shall select the sample from each asset category / Type & status (Sample size = 200 units for each DC covering both active and inactive assets). Physical verification (tracking testing) shall be conducted in presence of UIDAI representative monthly. The monthly report of the above activity shall be shared with UIDAI.	

	The automation between ITAM and IMAC to be implemented by MSIP to automate the real-time update of the asset location.
SLA Calculation	SLA formula= (Total Number of IT Assets successfully passed random test for the reporting month / Total Number of IT Assets on which random test was conducted for the reporting month) *100
Assumptions	Below mentioned are few scenarios when there could be deviation in compliance of ITAM database with Physical Verification of IT Assets. Assets in following categories would be excluded in the Sample for audit as approved by UIDAI on a case to case basis: • Bulk IT Asset movement • Asset in maintenance mode at the time of Random Asset Tracking Test/ Asset effected temporary due to ongoing incidents at time of test • Assets for which IMAC tickets is raised and Open
Measurement Interval	Monthly
Reporting Interval	Quarterly
Reports and data to be submitted	Quarterly SLA Report

1.5.20 To-Be SLA 29 - Asset Maintenance Cycle Adherence

Track / Category	Target	Applied Severity level
Inventory Management	100% compliance and no defaults in payments to OEMs for AMCs and warranties	None
	<100%	4
SLA Description	Asset Maintenance Cycle Adherence	
Definition of SLA	Asset Maintenance Cycle Adherence including timely payment for all AMCs and Warranties.	
Scope of SLA	The scope includes the tracking of asset AMC and warranty of MSIP supplied assets and for contracts formally assigned to MSIP by UIDAI for tracking. MSIP – Contract Manager (“CM”) will report to ADG-Tech, who assigns the contracts to CM track for monitoring. CM to maintain the AMC & Warranty tracker for the assets.	
Data Points for SLA Calculation	SLA will be computed based on formal submission of monthly Contract and Vendor performance report to UIDAI as per above scope. The relevant artefact, evidences for AMC and warranty payment will be submitted along with the report.	

SLA Calculation	For AMC & Warranty tracker for the assets supplied by MSIP, all MSIP contracts shall be compliant to Warranty or AMC as applicable for reporting month as per MSIP contract & vendor performance report for that month. Non-compliance to this shall be treated as SLA breach.
Assumptions	MSIP shall not be responsible for payments or any delay in payment to vendors for the contracts directly assigned to Vendors by UIDAI as the payment process is directly managed by UIDAI. MSIP scope is limited to tracking and reporting of the AMC and Warranty status to UIDAI in these cases. UIDAI shall formally communicate to MSIP for any contracts to be tracked by MSIP team. For any dependencies on UIDAI for AMC, MSIP will provide representation to UIDAI on case to case basis for SLA exemptions.
Measurement Interval	Monthly
Reporting Interval	Quarterly
Reports and data to be submitted	Tracker to be submitted as a part of monthly report on the first Monday of each month. Evidences for the AMC payment

1.5.21 To-Be SLA 30 - Patch regression testing

Track / Category	Target	Applied Severity level
Patch Management	100%	None
	<100%	4
SLA Description	Patch regression testing	
Definition of SLA	Metric: Regression Testing of the patch before patch upgrade on the system Formula: % Regression Testing Done = No. of Regression Testing of Patches / No. of Patch Upgrades	
Scope of SLA	The scope of this SLA shall include the OS patches, Database patches and patches of all products offered as PaaS services.	
Tool Used for SLA	Automated EMS implemented by MSIP	
Process to capture raw data for SLA calculations	Testing as per UIDAI's guidelines after any patch installation or upgrade	
SLA Calculation	Calculation Formula: % Regression Testing Done = No. of Regression Testing of Patches / No. of Patch Upgrades	
Assumptions	1: Valid for all UIDAI relevant patches only 2: Valid only for those specific instances where Staging instances are available	
Measurement Interval	Monthly	

Reporting Interval	Quarterly
Reports and data to be submitted	Quarterly SLA Report

1.5.22 To-Be SLA 31 - Patch Application

Track / Category	Target	Applied Severity level
Patch Management	100%	None
	<100%	4
SLA Description	Patch Application	
Definition of SLA	Metric: Time taken to apply patch after the patch is released for reported vulnerabilities. Maximum time allowed for patch application is 72 hours	
Scope of SLA	The scope considers all the patches (including OS, DB and applications) that are available for fixing the identified vulnerabilities	
Tool Used for SLA	Automated EMS implemented by MSIP	
Data Points for SLA Calculation	Start Time – reporting of the vulnerability, advisories from OEM on critical vulnerability, patches and date and time of release of patch by OEM, application vendor Artefacts of successful deployment of patches for closure date and time for SLA	
SLA Calculation	SLA calculation: No. of patches upgraded within 72 hours/Total no. of patches released for identified vulnerabilities The time taken by UIDAI for approvals and below dependencies shall be excluded from SLA calculations: 1. Dependencies on CAB, ECAB Approvals for deployment 2. Approval for restart of Services, Servers 3. Availability of Regression testing environment in case the OEM or Patch vendor mandates regression testing	
Assumptions	There may be cases wherein the SLA data is not captured in the respective automated, reporting tool due to unforeseen reasons. In such cases, appropriate artefacts, evidences will be considered for SLA calculation. There may be scenarios where MSIP may not be responsible for the breach of the SLA. These scenarios may include the dependencies beyond the control of MSIP, and / or managed by UIDAI or other ecosystem partners. Such cases will be exempted from SLA calculations provided appropriate artefacts or evidences are provided by MSIP.	
Measurement Interval	Monthly	
Reporting Interval	Quarterly	
Reports and data to be submitted	Quarterly SLA Report	

1.6 Information Security SLAs

The following SLAs shall be applicable for both As-Is and To-Be state.

1.6.1 To-Be SLA 32 - Vulnerability assessment

SLA Description	Vulnerability Assessment
Scope of SLA	Applies to all Active Devices at UIDAI- physical, virtual machine, containers, Management IPs
Tools, methods or scripts to be used to measure SLA	Vulnerability Management tools, Service Request tool, GRC Archer and Asset Management used for SLA Management
Process to capture data or measure the parameter	In order to measure Security posture of UIDAI, it is not sufficient to have the systems up and running. The MSIP has to use the vulnerability management tools to scan systems or devices active in UIDAI as per asset management tool. <u>Measurement of the SLA</u> This SLA shall be measured as follows. Vulnerability assessment shall be conducted every quarter on all the active assets and the finding of assessment shall be uploaded on GRC Archer (at present) or any other tool as prescribed by UIDAI with system or device priority from asset management tool and allocate the vulnerabilities for resolution to concerned asset owners. The vulnerabilities need to be addressed within agreed timeframe. Note: Tools may categorize the vulnerabilities as (High/Critical, Medium, Low) or equivalent: ▪ Vulnerabilities with High priority should be addressed within one week from the date of reporting or one week from the date of detection, whichever is earlier ▪ Vulnerabilities with Medium priority should be addressed within 30 days from the date of reporting ▪ Rest of the vulnerabilities should be addressed within 90 days from the date of reporting ▪ Exceptions have to be recommended for approval by Business owner approval and ADG IS security. All the Exceptions would be approved by CEO. ▪ No Vulnerability should be carried over without closure or an Exception on closure as per the time line below: a. In case of High Vulnerabilities- Exception should be raised within one week of reporting/detection for the maximum period of 30 days. b. In Case of Medium/Low Vulnerabilities- Exception should be raised within stipulated closure timelines of the vulnerability for the maximum period of 90 days.

SLA Description	Vulnerability Assessment
	c. Exception for the same device, software, appliance, system or application should not be sought for more than two times unless there is an OEM dependency. - It is to be ensured that a vulnerability is resolved, the same should not be identified in any subsequent assessments in the same device, software, appliance, system or application for whatsoever reason, ever. If identified, such recurrence will be considered as a breach of this SLA and LD will be applicable for all the previous quarters since when the vulnerability was claimed close. # Repetition of vulnerability due to any intermediate changes like system upgrades, patches or OS changes etc. in the same system/application/device etc. will be considered as repeated as the scanning has to be carried out after any changes in the subject of vulnerability.
SLA calculation	Formula: Compliance % for High vulnerabilities = (Number of High vulnerabilities generated during an assessment and closed within 1 week of date of logging/Number of previously reported High vulnerabilities which are open without exception approval + Number of High vulnerabilities reported during assessment– Number of such vulnerabilities for which approvals for the exceptions raised have been obtained) AND Compliance % for Medium vulnerabilities = (Number of Medium or Major vulnerabilities reported during an assessment and closed within 30 days of date of logging/ Number of previously reported Medium vulnerabilities which are open without exception approval + Number of Medium vulnerabilities reported during an assessment – Number of such vulnerabilities for which approvals for the exceptions raised have been obtained) AND Compliance % for other vulnerabilities = (Number of other vulnerabilities reported during an assessment and closed within 90 days of date of logging/ (Number of previously reported other vulnerabilities which are open without exception approval + Number of other vulnerabilities reported during an assessment – Number of such vulnerabilities for which approvals for the exceptions raised have been obtained) AND No. of repeat vulnerabilities which were claimed closed earlier=0.
SLA target	Compliance = 100% for all categories of vulnerabilities AND Recurrence of vulnerability previously identified and closed in same device, software, appliance, system or application = NIL
Severity level for computation of Liquidated Damages	2 for medium & low vulnerabilities 4 for high & Repeated Vulnerabilities of high, medium or other categories

SLA Description	Vulnerability Assessment
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics in case the same functionalities are not available out of the box in standard tool (GRC Archer at present). Exceptions cannot be sought for more than two times for same device unless there is an OEM dependency. MSIP will have to provide the artefact from the OEM to confirm the dependency. Verification of compliance may be carried out by an internal or third-party auditor
Measurement interval	Monthly
Reporting interval	Quarterly
Reports or Data to be submitted, if any	Reporting should be via an automated SLA dashboard. At a minimum the following reports should be submitted. Further reports should be customizable by UIDAI users in SLA reporting tool as required. 1. As is Vulnerability Assessment landscape report 2. Compliance Report for high vulnerabilities– First Week 3. Compliance Report for medium vulnerabilities– 30 days 4. Vulnerability Compliance Report from– 90 days 5. Ad-HOC compliance report as per UIDAI requirement
Illustration for LD computation	For a given quarter, assume the following scenarios. Scenario 1: SLA targets met LD applicable: Nil Scenario 2: i. Compliance % of medium & low vulnerabilities = 95% ii. Compliance % for high and repeated vulnerabilities = 100% LD applicable: 0.2% for severity level 2 for non-compliance with SLA target for medium and low vulnerabilities

1.6.2 To-Be SLA 33 - Penetration Testing (PT)

SLA Description	Penetration Testing Assessment
Scope of SLA	Applies to UIDAI All Active Portals (Internal/External)
Tools, methods or scripts to be used to measure SLA	Penetration testing tools, Service Request tool, GRC and Asset Management used for SLA Management
Process to capture data or measure the parameter	In order to measure Security posture of UIDAI, it is not sufficient to have the systems are up and running. The MSIP has to use the Penetration testing tools to scan all active portals. <u>Measurement of the SLA</u> This SLA shall be measured as follows. Penetration Testing(PT) should be conducted on bi-annual basis on all the UIDAI active portals. The findings

SLA Description	Penetration Testing Assessment
	of the PT shall be uploaded and the closure status be updated on GRC Archer application (at present) for tracking and monitoring purpose. The identified PT findings needs to be addressed within agreed timeframe: Note: Tools may categorize the vulnerabilities as (High/Critical, Medium, Low) or equivalent. ▪ PT findings with High severity should be addressed within one week from the date of reporting ▪ PT findings with Medium severity should be addressed within 30 days from the date of reporting ▪ Rest of the PT findings should be addressed within 90 days from the date of reporting ▪ Exceptions have to be recommended for approval by Business owner approval and ADG IS security. All the Exceptions would be approved by CEO. ▪ No PT findings should be carried over without closure or an Exception on closure as per the time line below: a. In Case of High PT findings, Exception should be raised within one week of reporting for the maximum period of 30 days. b. In Case of Medium/Low PT findings, Exception should be raised within stipulated closure timelines of the PT findings for the maximum period of 90 days. ▪ Exception for the same device, software, appliance, system or application should not be sought for more than two times unless there is an OEM dependency. ▪ It is to be ensured that a PT finding is resolved, the same should not be identified in any subsequent assessments in the same portal for whatsoever reason. If identified such recurrence will be considered as a breach of this SLA and LD will be applicable for all the previous quarters since when the vulnerability was claimed close. ▪ *Repetition of PT findings due to any intermediate changes like system upgrades, patches or OS changes etc. in the portal will be considered as repeated, as thorough testing should be carried out after any changes in the portal.
SLA calculation	Formula: Compliance % for High PT findings = (Number of High PT findings generated during an assessment and closed within 1 week of date of reporting/Number of previously reported High PT findings which are open without exceptions + Number of High PT findings reported during assessment– Number of such PT findings for which approvals for the exceptions raised have been obtained) AND Compliance % for Medium PT findings = (Number of Medium or Major PT findings reported during an assessment and closed within 30 days of date of Reporting/ Number of previously reported Medium PT findings which

SLA Description	Penetration Testing Assessment
	are open without exceptions + Number of Medium PT findings reported during an assessment – Number of such PT findings for which approvals for the exceptions raised have been obtained) Compliance % for other PT findings = (Number of other PT findings reported during an assessment and closed within 90 days of date of reporting/ (Number of previously reported other PT findings which are open without exceptions + Number of other PT findings reported during an assessment – Number of such PT findings for which approvals for the exceptions raised have been obtained) AND No. of repeat PT findings which were claimed closed earlier=0. *In case of any repeated PT finding, LD will apply for all the quarters since the time it was claimed close.
SLA target	Compliance = 100% for all categories of PT findings AND Recurrence of PT findings previously identified and closed in same portal = NIL
Severity level for computation of Liquidated Damages	2- External Low, 2- Internals (medium and Low) 2-Repeated low 3- Internal high, 3- External (high and medium) 3- Repeated High Medium & PT findings
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard tool. Exceptions cannot be sought for more than two times for same device unless there is an OEM dependency. MSIP will have to provide the certificate from the OEM to confirm the dependency. Verification of compliance may be carried out by an internal or third-party auditor
Measurement interval	Bi-annual
Reporting interval	Bi-annual
Reports or Data to be submitted, if any	Reporting should be via an automated SLA dashboard. At a minimum the following reports should be submitted. Further reports should be customizable by UIDAI users in SLA reporting tool as required. - As is PT findings landscape report - PT findings Compliance Report from – First Week

SLA Description	Penetration Testing Assessment		
	3. PT findings Compliance Report from– 30 days 4. PT findings Compliance Report from– 90 days 5. Ad-HOC compliance report as per UIDAI requirement		
Illustration for LD computation	For a given quarter, assume the following scenarios.		
	Scenario 1: SLA targets met LD applicable: Nil		
	Scenario 2:		
	SLA Target	Achievement	LD applicable %
	Compliance % for External Low, Internals (medium and Low) and Repeated low	85%	0.2% corresponding to severity level 2
Compliance % for Internal high, External (high and medium) And Repeated High Medium & PT findings	90%	0.3% corresponding to severity level 3	
LD applicable for the quarter		= (0.2% + 0.3%)/2 = 0.25%	

1.6.3 To-Be SLA 34 - Endpoint Security Assessment (AV & Malware)

SLA Description	End point security assessment
Scope of SLA	Applies to all Active Devices at UIDAI- physical, virtual machine, containers, Management IPs. The MSIP has to conduct an assessment every week as per the agreed schedule. MSIP has to support ad-hoc or third-party assessment conducted by UIDAI as per defined priorities in resolving the AV and malware issues.
Tools, methods or scripts to be used to measure SLA	End Point Security Management tools, Asset Management and Service Request tool used for SLA Management.
Process to capture data or measure the parameter	- The MSIP has to use the Antivirus and malware tools to scan systems and devices active in UIDAI and Data centre as per asset management tool. - This operation should be conducted at regular intervals and AV and Malware report should be submitted to UIDAI weekly. - Tool used must provide complete coverage, percentage of coverage, scanning and cleaning level and update. - In case of recurring instances, RCA should be performed and infection should be resolved permanently. RCA to be conducted within 24 Hrs of detection and fixed to be provided within 48 hrs. These timelines will be measured based on the Incident lifecycle.

SLA Description	End point security assessment																																		
	e. Repetition of Instances, if any should be reported in the weekly report and consolidated quarterly SLA report.																																		
SLA calculation	Formula: Scan Coverage (%) = Number of active devices scanned/ Number of active devices Scheduled scan (%)= Number of active devices had a weekly scan/ Number of active devices RCA Completion and Fix % = Number of devices on which RCA was done and fixed/Number of devices or systems on which RCA was due as per assessment Number of repetitions= No of such instances found repeated on the same device, which has been claimed cleaned ever. *RCA is mandatory in case same virus or malware occurs more than once within a month) on the same system. *RCA is mandatory in case same virus/malware has infected more than 5 systems.																																		
SLA target and Severity level for computation of Liquidated Damages	<table><tr><th>Scan Coverage</th><th>Scheduled Scan</th><th>No. of repeated malware attacks</th><th>RCA Conducted and Fixed</th><th>Severity Level</th></tr><tr><td>>=95%</td><td>>=95%</td><td><=2</td><td>100%</td><td>None</td></tr><tr><td><95% and >=90%</td><td><95% and >=90%</td><td>>2 and <=4</td><td><100% and >=95%</td><td>3</td></tr><tr><td><90% and >=85%</td><td><90% and >=85%</td><td>>6 and <=8</td><td><90% and >=85%</td><td>4</td></tr><tr><td><85% and >=80%</td><td><85% and >=80%</td><td>>8 and <=10</td><td><<85% and >=80%</td><td>5</td></tr><tr><td><80%</td><td><80%</td><td>>10</td><td><80%</td><td>6</td></tr></table>					Scan Coverage	Scheduled Scan	No. of repeated malware attacks	RCA Conducted and Fixed	Severity Level	>=95%	>=95%	<=2	100%	None	<95% and >=90%	<95% and >=90%	>2 and <=4	<100% and >=95%	3	<90% and >=85%	<90% and >=85%	>6 and <=8	<90% and >=85%	4	<85% and >=80%	<85% and >=80%	>8 and <=10	<<85% and >=80%	5	<80%	<80%	>10	<80%	6
Scan Coverage	Scheduled Scan	No. of repeated malware attacks	RCA Conducted and Fixed	Severity Level																															
>=95%	>=95%	<=2	100%	None																															
<95% and >=90%	<95% and >=90%	>2 and <=4	<100% and >=95%	3																															
<90% and >=85%	<90% and >=85%	>6 and <=8	<90% and >=85%	4																															
<85% and >=80%	<85% and >=80%	>8 and <=10	<<85% and >=80%	5																															
<80%	<80%	>10	<80%	6																															
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard tool. Exceptions cannot be sought unless there is an OEM dependency. MSIP will have to provide the artefact from the OEM to confirm the dependency. Verification of compliance may be carried out by an internal or third-party auditor																																		
Measurement interval	Monthly																																		
Reporting interval	Quarterly																																		

SLA Description	End point security assessment		
Reports or Data to be submitted, if any	Automated reporting via SLA reporting tools. - AS-IS Antivirus and Malware landscape report – Thrice a day - Consolidated AS-IS Antivirus and Malware landscape report – Weekly - Anti-Virus and Malware Compliance Report – Monthly - Recurring Virus Hit Count - RCA completion and fix implementation report		
Illustration of LD computation	SLA target	Achievement (Qtr.)	LD applicable %
	Scan coverage	90%	0.3%
	Scheduled scan	97%	Nil
	No. of maximum repetition of malware attack	3	0.3%
	Root Cause Analysis Conducted and Fixed	98%	0.3%
	LD applicable for the quarter		$= (0.3\% + 0\% + 0.3\% + 0.3\%)/4 = 0.23\%$

1.6.4 To-Be SLA 35 - Coverage of system hygiene

SLA Description	Coverage of System Hygiene
Scope of SLA	Applies to all active standalone physical server, Appliance, virtual machine, container, portal or service (All IP Address Active)
Tools, methods or scripts to be used to measure SLA	EMS tools to be implemented by MSIP – a custom monitoring script shall be executed at least once a month from the SLA monitoring server. Definition of hygiene: for given below assets the respective security requirements are to be implemented: - For Endpoint systems (Desktops/Laptops/Virtual Desktops)-DLP/AV/AD/OS updates - For Servers (Physical and Virtual): OS updates/AV/LDAP/AD Integration - Perimeter devices: Firewall/Access Routers/Switches/Load Balancers/Security Devices etc., OS Updates/ Latest signature(N-1) - Tools (SIEM/GRC Archer/QUALYS/ ENS/IAM etc.)- Latest patch
Process to capture data or measure the parameter	The MSIP has to measure the coverage of System Hygiene at regular interval for all systems against a standard template of compliance, configuration to conformation against all active assets in UIDAI (HQ, ROs and Tech Center) and CIDR.
SLA calculation	Formula: $\% \text{ Coverage of Hygiene} = (\text{Number of updated/configured active devices with DLP and other security tools} / \text{Total number of active devices})$

SLA Description	Coverage of System Hygiene		
SLA target and Severity Level for computation of Liquidated Damages	Coverage of Hygiene		Severity Level
	>=95%		None
	<95% and >= 90%		1
	<90% and >= 85%		2
	<85% and >=80%		3
	<80% and >=75%		4
	<75% and >=70%		5
	<65% and >=60%		6
	<60% and >=55%		7
	<55% and >=50%		8
	<50% and >=40%		9
	<40%		10
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard tool. Compliance templates can be reviewed by UIDAI at regular interval for addition or deletion of compliance requirements to meet business requirements. Exceptions cannot be sought for more than two times for same device unless there is an OEM dependency. MSIP will have to provide the artefact from the OEM to confirm the dependency. Verification of process compliance may be carried out by an internal or third-party auditor		
Measurement interval	Monthly		
Reporting interval	Quarterly		
Reports or Data to be submitted, if any	Automated SLA dashboard in EMS tool and conformation to various security systems deployed - As is compliance Assessment landscape report - Coverage of compliance report domain wise (like AV, DLP, etc.) – Monthly - AD-HOC compliance report - Security devices compliance report - Compliance exceptions report for previous quarter for validation and comparison		
Illustration of LD computation	SLA target	Achievement	LD applicable %
	% Coverage of Hygiene	Scenario 1: 90%	0.1%
	% Coverage of Hygiene	Scenario 2: 45%	0.9%

1.6.5 To-Be SLA 36 - Access Controls and Violations

SLA Description	Access Controls and Violations
Scope of SLA	Applies to UIDAI all active standalone physical server, Appliance, virtual machine, container or service. Access violation will include but not be limited to the following: - Direct access to servers and critical devices at any time. Direct access means accessing the devices bypassing, IAM or Active Directory/LDAP. - Root access in servers without approval - Login without 2-factor authentication in critical devices **Critical devices in the above context includes Switches, firewalls, routers, EPOS., IPS, Load Balancer etc.
Tools, methods or scripts to be used to measure SLA	IAM, SIEM, Syslog and EMS tools to be implemented by MSIP – a custom monitoring script shall be executed at least once every minute from the SLA monitoring service. The MSIP has to conduct a formal assessment every quarter as per defined schedule. MSIP has to support ad-hoc or third-party assessment conducted by UIDAI as per defined priorities in resolving the access violations.
Process to capture data or measure the parameter	- The MSIP has to measure and report the access violations and access management of all critical systems active in UIDAI and CIDR and has to submit log report from SIEM or similar tool monthly. - Exceptions cannot be sought for more than two times for same device unless there is an OEM dependency. - MSIP will have to provide the artefact from the OEM to confirm the dependency. - Verification of all the process compliance by an internal or third-party auditor.
SLA calculation and target	No of violations for Root access without approval = NIL AND Login without 2-factor Target =NIL AND Repeat access violations = NIL
Severity level for computation of Liquidated Damages	4
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard EMS tools.
Measurement interval	Monthly



SLA Description	Access Controls and Violations
Reporting interval	Quarterly
Reports or Data to be submitted, if any	SLA reporting should be through automated tool. It should report the following: i. Number of Privileged access violations report ii. Separation of duty violations report iii. Repeat access violations iv. Access control and violations exceptions report for previous quarter for validation and comparison

1.6.6 To-Be SLA 37 - Incident Response Time

SLA Description	Incident Response Time																																								
Scope of SLA	Applies to all Incidents																																								
Tools, methods or scripts to be used to measure SLA	EMS - Service Request Management tools to be implemented by MSIP																																								
Process to capture data or measure the parameter	In order to measure the Service request's quality, it is important to monitor the service request time, service request response time, service request resolution or escalation reports. The MSIP can either use customized scripts or customization of tools necessary for measuring the SLAs. Once a ticket/incident is logged through one of the agreed channels, the priorities have to be set using agreed channels for tickets based on UIDAI requirements to meet and tune the business objectives. Measurement of SLA Response Time is measured from the time of creation of the ticket until the "In Progress" status update. <table><tr><th>Priority</th><th>Response time</th></tr><tr><td>P1</td><td>15 minutes</td></tr><tr><td>P2</td><td>60 minutes</td></tr><tr><td>P3</td><td>2 hours</td></tr><tr><td>P4 - Service Request</td><td>1 Day</td></tr></table> The priority of incidents shall be measured as per the table below. <table><tr><th colspan="5">Asset Criticality/ Urgency</th></tr><tr><th>Impact</th><th>High</th><th>Medium</th><th>Low</th><th>Service Request</th></tr><tr><td>Entire organization</td><td>P1</td><td>P1</td><td>P3</td><td>P4</td></tr><tr><td>Large Group (>20 persons)</td><td>P1</td><td>P2</td><td>P3</td><td>P4</td></tr><tr><td>Small Group (=<10 persons)</td><td>P2</td><td>P3</td><td>P3</td><td>P4</td></tr><tr><td>One Person</td><td>P2</td><td>P3</td><td>P3</td><td>P4</td></tr></table>	Priority	Response time	P1	15 minutes	P2	60 minutes	P3	2 hours	P4 - Service Request	1 Day	Asset Criticality/ Urgency					Impact	High	Medium	Low	Service Request	Entire organization	P1	P1	P3	P4	Large Group (>20 persons)	P1	P2	P3	P4	Small Group (=<10 persons)	P2	P3	P3	P4	One Person	P2	P3	P3	P4
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P4 - Service Request	1 Day																																								
Asset Criticality/ Urgency																																									
Impact	High	Medium	Low	Service Request																																					
Entire organization	P1	P1	P3	P4																																					
Large Group (>20 persons)	P1	P2	P3	P4																																					
Small Group (=<10 persons)	P2	P3	P3	P4																																					
One Person	P2	P3	P3	P4																																					
SLA calculation	Formula: Response time % for P1 tickets = Number of P1 tickets opened during the period and for which response is provided within defined timelines during the period / Number of P1 tickets opened during the period																																								

SLA Description	Incident Response Time		
	Response time % for P2 tickets = Number of P2 tickets opened during the period and for which response is provided within defined timelines during the period / Number of P2 tickets opened during the period Response time % for P3 tickets = Number of P3 tickets opened during the period and for which response is provided within defined timelines during the period / Number of P3 tickets opened during the period Response time % for P4 tickets = Number of P4 tickets opened during the period and for which response is provided within defined timelines during the period / Number of P4 tickets opened during the period		
SLA target	Response Time compliance =100%		
Severity level for computation of Liquidated Damages	2		
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard Service Request tools.		
Measurement interval	Monthly		
Reporting interval	Quarterly		
Reports or Data to be submitted, if any	Automated SLA dashboard in Service Request tools but not limiting to following reports can be customizable as required. Ticket Response Report ticket Request Work "In Progress" Report		
Illustration of LD computation	SLA target	Achievement	LD applicable %
	Response Time compliance =100%	95%	0.2% corresponding to severity level 2

1.6.7 To-Be SLA 38 - Incident Resolution Time

SLA Description	Incident Resolution Time																																								
Scope of SLA	Applies to all incidents																																								
Tools, methods or scripts to be used to measure SLA	EMS - Service Request Management tools to be implemented by MSIP																																								
Process to capture data or measure the parameter	In order to measure the incident management process, it is important to monitor the incident response time. Incidents are automatically created and assigned to a group from SIEM, Syslog, Firewall, etc. The MSIP can either use customized scripts or customization of tools necessary for measuring the SLAs. Once an incident is logged through one of the agreed channels, the priorities have to be set using agreed channels for tickets based on UIDAI requirements to meet and tune the business objectives. Measurement of SLA Response time is the time from incident creation until the "Work in Progress" status update with assignment by MSIP. Resolution Time is the time from incident creation until the "Resolved" status update. <table><tr><th>Priority</th><th>Resolution time</th></tr><tr><td>P1</td><td>8 Hours</td></tr><tr><td>P2</td><td>2 Days</td></tr><tr><td>P3</td><td>5 days</td></tr><tr><td>P4 - Service Request</td><td>15 days</td></tr></table> The priority of incidents shall be measured as per the table below. <table><tr><th colspan="5">Asset Criticality/ Urgency</th></tr><tr><th>Impact</th><th>High</th><th>Medium</th><th>Low</th><th>Service Request</th></tr><tr><td>Entire organization</td><td>P1</td><td>P1</td><td>P3</td><td>P4</td></tr><tr><td>Large Group (>20 persons)</td><td>P1</td><td>P2</td><td>P3</td><td>P4</td></tr><tr><td>Small Group (=<10 persons)</td><td>P2</td><td>P3</td><td>P3</td><td>P4</td></tr><tr><td>One Person</td><td>P2</td><td>P3</td><td>P3</td><td>P4</td></tr></table>	Priority	Resolution time	P1	8 Hours	P2	2 Days	P3	5 days	P4 - Service Request	15 days	Asset Criticality/ Urgency					Impact	High	Medium	Low	Service Request	Entire organization	P1	P1	P3	P4	Large Group (>20 persons)	P1	P2	P3	P4	Small Group (=<10 persons)	P2	P3	P3	P4	One Person	P2	P3	P3	P4
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Asset Criticality/ Urgency																																									
Impact	High	Medium	Low	Service Request																																					
Entire organization	P1	P1	P3	P4																																					
Large Group (>20 persons)	P1	P2	P3	P4																																					
Small Group (=<10 persons)	P2	P3	P3	P4																																					
One Person	P2	P3	P3	P4																																					
SLA calculation	Formula: Resolution time % for P1 tickets = Number of P1 tickets opened during the period and for which resolution is provided within defined timelines during the period / Number of P1 tickets opened during the period																																								

	Resolution time % for P2 tickets = Number of P2 tickets opened during the period and for which resolution is provided within defined timelines during the period / Number of P2 tickets opened during the period Resolution time % for P3 tickets = Number of P3 tickets opened during the period and for which resolution is provided within defined timelines during the period / Number of P3 tickets opened during the period Resolution time % for P4 tickets = Number of P4 tickets opened during the period and for which resolution is provided within defined timelines during the period / Number of P4 tickets opened during the period		
SLA target	Resolution Time % = 100%		
Severity level for computation of Liquidated Damages	4		
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard Service Request tools.		
Measurement interval	Monthly		
Reporting interval	Quarterly		
Reports or Data to be submitted, if any	Automated SLA dashboard in Service Request tools but not limiting to following reports can be customizable as required. 1. Ticket Resolution Report 2. Ticket Request Work "In Progress" Report 3. Ticket average resolution report 4. Ticket number of duplicates 5. % major tickets converted as Problem 6. % repeat unknown cause of tickets 7. No of tickets with priorities modified during their course of resolution report 8. RCA of P1 and P2 resolved.		
Illustration of LD computation	SLA target	Achievement	LD applicable %
	Resolution Time compliance =100%	95%	0.4% corresponding to severity level 4

1.6.8 To-Be SLA 39 - Problem Ticket Management

SLA Description	Problem Ticket Management
Scope of SLA	Applies to all Problem tickets. In this context, a Problem is defined as follows: if same issue reoccurs ever (during the entire contract period) for 5 or more times on same device or system or application, a Problem ticket has to be created and RCA (Actual Root Cause behind the issue) of all such tickets has to be submitted as per timeline (defined below) In addition, MSIP has to conduct trend analysis of historic data and identify recurring issue categories at asset level, convert such incidents into problems and follow the problem management lifecycle to arrive at a permanent fix.
Tools, methods or scripts to be used to measure SLA	Security Service Request Management tools to be implemented by MSIP
Process to capture data or measure the parameter	In order to measure the problem management effectiveness, it is important to monitor the response time, and recurrence. The MSIP can either use customized scripts or customization of tools necessary for measuring the SLAs. <u>Measurement of SLA</u> Definition: Resolution Time is the time from service request creation until the "Resolved" status update *Problem Ticket should be opened within 7 days of meeting the criteria of identifying a problem. RCA should be submitted within 30 days of opening the Problem ticket. *A Problem along with RCA should be resolved within 90 Days of opening a problem ticket. Exceptions have to be recommended for approval by Business owner approval and ADG IS security. All the Exceptions would be approved by CEO UIDAI.
SLA calculation	Formula: Failure in meeting the timeline for any Problem ticket creation OR Failure in meeting the timeline for performing RCA on all the Problem tickets created Failure in meeting the timeline for performing Closure of all the Problem tickets created *If a scenario for a 'Problem' as mentioned in this SLA is identified by UIDAI or third party and for which a Problem ticket was not created by MSIP. In Such a scenario it would be considered as

SLA Description	Problem Ticket Management			
	“Breach of SLA” & considered not met since from the quarter this ticket was supposed to be created.			
SLA target	1. Creation Time =100% 2. RCA performed = 100% 3. Resolved % = 100%			
Severity level for computation of Liquidated Damages	Creation time	RCA performed	Resolved %	Severity Level
	100%	100%	100%	None
	<100% and >=95%	<100% and >=95%	<100% and >=95%	2
	<95% and >=90%	<95% and >=90%	<95% and >=90%	4
	<90% and >=85%	<90% and >=85%	<90% and >=85%	6
	<85% and >=80%	<85% and >=80%	<85% and >=80%	8
	<80%	<80%	<80%	10
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard Service Request tools.			
Measurement interval	Monthly			
Reporting interval	Quarterly			
Reports or Data to be submitted, if any	Reporting through Automated SLA dashboard. Following reports/metrics should be reported at a minimum. 1. Problem Creation Report- Monthly 2. Problem Response Report-Monthly 3. Problem RCAs Report-Monthly 4. Problem Resolution Report- Monthly *Known Error Data-Base has to be maintained *Known Error Records should be updated against each identified problem. *UIDAI has authority to convert appropriate high priority tickets into Problem Category			
Illustration of LD computation	SLA target	Achievement (Qtr.)	LD applicable %	
	Creation time =100%	95%	0.2%	
	RCA performed = 100%	98%	0.2%	
	Resolved %	100%	Nil	
	LD applicable		$(0.2\% + 0.2\% + 0\%)/3 = 0.13\%$	

1.6.9 To-Be SLA 40 - Third Party Assessments

SLA Description	Third Party Assessments
Scope of SLA	Applies to the findings of Assessments conducted by GRCP or any third party against Processes (like Incident management, Risk Assessment, BCP etc.) applications, tools (e.g. GRC Archer, DLP, IAM, SIEM etc.) /processes / tools)/facilities etc.
Tools, methods or scripts to be used to measure SLA	Checklists, templates and reports for detailed assessment findings
Process to capture data or measure the parameter	In order to measure the success of findings of all assessments MSIP should address the findings/gaps for closure with relevant artefact in support <u>Measurement of the assessment</u> Findings assigned to the concerned stakeholder within MSIP has to provide the closure/improvement plan of findings/gaps identified to UIDAI within 60 days of receipt of said report to MSIP. 1. Acknowledgement with-in 30 days of receipt – Should be 100% 2. 50% closure of identified observation Within 60 days of receipt 3. 100% closure with-in 120 days against identified observations
SLA calculation	Formula: Compliance % on response= Closure or improvement plan submitted for no. of findings /Total no. of findings for MSIP Compliance % on quality of closure or improvement plan= No. of closures or improvement plans accepted by UIDAI/ Total number of findings for MSIP
SLA target	Compliance on Response=100% AND Compliance % on quality of closure or improvement =>50%
Severity level for computation of Liquidated Damages	2
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard Service Request tools. Verification of all the process compliance by an internal or third-party auditor. Exceptions have to be recommended for approval by Business owner approval and ADG IS security. All the Exceptions would be approved by CEO
Measurement interval	Quarterly
Reporting interval	Quarterly



SLA Description	Third Party Assessments		
Reports or Data to be submitted, if any	Response to findings Closure/ improvement plans report		
Illustration of LD computation	SLA target	Achievement	LD applicable %
	Compliance on Response=100%	95%	0.2% corresponding to severity level 2
	Compliance % on quality of closure or improvement =>50%	98%	Nil
	LD applicable		= (0.2%+0%)/2=0.1%

1.6.10 To-Be SLA 41 - CRM Fraud Management System (“FMS”) Tickets Response

SLA Description	CRM (FMS) Tickets Response																								
Scope of SLA	Applies to all citizen service requests received through UIDAI helpline or CRM tool requests or UIDAI ad-hoc cases allocated to FMS queue. FMS is expected to acknowledge to the tickets/ad-hoc cases through emails allocated to FMS.																								
Tools, methods or scripts to be used to measure SLA	Service Request tool, CRM, UIDAI helpline, emails from UIDAI																								
Process to capture data or measure the parameter	Measurement of SR Response SLA SLA (parameters) – Acknowledgement of all cases within 2 working days from the case assignment dates in FMS. *Response/acknowledgment in this context is the time recorded in ticketing system																								
SLA calculation	Formula: Compliance Ticket Response (%) = (Number of tickets acknowledged in defined timelines / (Total Number of tickets allocated +received through email and Ad-Hoc etc.))																								
SLA target and Severity level for computation of Liquidated Damages	<table> <tr> <th>Ticket Response %</th><th>Severity Level</th></tr> <tr> <td>100%</td><td>None</td></tr> <tr> <td><100% and >=98%</td><td>1</td></tr> <tr> <td><98% and >=96%</td><td>2</td></tr> <tr> <td><96% and >=94%</td><td>3</td></tr> <tr> <td><94% and >=92%</td><td>4</td></tr> <tr> <td><92% and >=90%</td><td>5</td></tr> <tr> <td><90% and >=85%</td><td>6</td></tr> <tr> <td><85% and >=80%</td><td>7</td></tr> <tr> <td><80% and >=75%</td><td>8</td></tr> <tr> <td><75% and >=70%</td><td>9</td></tr> <tr> <td><70%</td><td>10</td></tr> </table>	Ticket Response %	Severity Level	100%	None	<100% and >=98%	1	<98% and >=96%	2	<96% and >=94%	3	<94% and >=92%	4	<92% and >=90%	5	<90% and >=85%	6	<85% and >=80%	7	<80% and >=75%	8	<75% and >=70%	9	<70%	10
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<85% and >=80%	7																								
<80% and >=75%	8																								
<75% and >=70%	9																								
<70%	10																								
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard Service Request tools.																								
Measurement interval	Monthly																								
Reporting interval	Quarterly																								
Reports or Data to be submitted, if any	Automated SLA dashboard in Service Request tool/CRM/FMS console but not limiting to following reports can be customizable as required. Ticket/ Inspection Response Report Ticket/ Inspection Resolution Report																								

SLA Description	CRM (FMS) Tickets Response		
Illustration of LD computation	SLA target	Achievement	LD applicable %
	Ticket response =100%	95%	0.3%

1.6.11 To-Be SLA 42 - CRM Fraud Management System (“FMS”) Tickets Resolution

SLA Description	CRM (FMS) Tickets																								
Scope of SLA	Applies to all citizen service requests received through UIDAI helpline or CRM tool requests or UIDAI ad-hoc cases allocated to FMS queue. FMS is expected to resolve these issues and provide responses to the issues highlighted by resident. This may involve speaking to the residents / data review and retrieval.																								
Tools, methods or scripts to be used to measure SLA	Service Request tool, CRM, UIDAI helpline																								
Process to capture data or measure the parameter	Once a ticket/incident is logged through one of the agreed channels, the priorities are set using agreed channels for requests allocated to FMS queue. Measurement of SR Resolution SLA SLA (parameters) – Closure of all cases within 15 working days from the case assignment dates in FMS (excluding justifiable & approved dependency at UIDAI's discretion, if any). Exception process as given in above SLAs, ADG approval accepted																								
SLA calculation	Formula: Compliance Ticket Resolution (%) = (Number of tickets resolved in defined timelines+ approved exceptions tickets) / (Total Number of tickets allocated – tickets raised for exception approval)																								
SLA target	<table> <tr> <th>Ticket Response %</th><th>Severity Level</th></tr> <tr> <td>100%</td><td>None</td></tr> <tr> <td><100% and >=98%</td><td>1</td></tr> <tr> <td><98% and >=96%</td><td>2</td></tr> <tr> <td><96% and >=94%</td><td>3</td></tr> <tr> <td><94% and >=92%</td><td>4</td></tr> <tr> <td><92% and >=90%</td><td>5</td></tr> <tr> <td><90% and >=85%</td><td>6</td></tr> <tr> <td><85% and >=80%</td><td>7</td></tr> <tr> <td><80% and >=75%</td><td>8</td></tr> <tr> <td><75% and >=70%</td><td>9</td></tr> <tr> <td><70%</td><td>10</td></tr> </table>	Ticket Response %	Severity Level	100%	None	<100% and >=98%	1	<98% and >=96%	2	<96% and >=94%	3	<94% and >=92%	4	<92% and >=90%	5	<90% and >=85%	6	<85% and >=80%	7	<80% and >=75%	8	<75% and >=70%	9	<70%	10
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SLA Description	CRM (FMS) Tickets		
Assumptions, if any	The MSIP should implement custom scripts to enable the measurement of various metrics when such functionalities are not available out of the box in standard Service Request tools.		
Measurement interval	Fortnightly		
Reporting interval	Quarterly (SLA computed will be simple average of all fortnightly measurements)		
Reports or Data to be submitted, if any	Automated SLA dashboard in Service Request tool/CRM/FMS console but not limiting to following reports can be customizable as required. Ticket/ Inspection Response Report Ticket/ Inspection Resolution Report		
Illustration of LD computation	SLA target	Achievement	LD applicable %
	Compliance ticket resolution =100%	95%	0.3%

1.6.12 To-Be SLA 43 - Compliance – Process documents maintenance

SLA Description	Process Adequacy, Review and Control
Scope of SLA	Applies to all Policies, Standard Operating Procedures, workflows, list of required Artefacts, Process Documents and any mandatory document required for ISMS audits that is pertaining to IS Cluster Operations.
Tools, methods or scripts to be used to measure SLA	Approved documents with updated version control history should be uploaded on KM portal.
Process to capture data or measure the parameter	Measurement of SLA - MSIP Process Owners should ensure to review and update all the documents with version control as per the planned schedule below OR before the schedule may be required if significant changes occur to ensure its continuing suitability, adequacy, adaptivity and effectiveness of the mentioned documents before *Approved document should be uploaded on the KM portal/SharePoint in designated folders. *Any document for approval of UIDAI should be submitted at least 10 days prior to scheduled submission date - ISMS Document: Annual - Process manual (e.g. Incident management, Change management, problem management and event management etc.): Half yearly - Tool based Standard Operations Procedure document (e.g. SIEM, DLP, SOC operations, fine-tuning of rules/alerts etc.): Half yearly - Mandatory regulatory documents: as and when available.

SLA Description	Process Adequacy, Review and Control		
SLA calculation	Formula: All the identified and approved documents should be uploaded on KM Portal/SharePoint in designated folders on or before Scheduled review date (1 st Feb and 1 st Aug of every year). In addition to the one-time application of LD as per the Severity level associated with the target, an additional 0.01% of LD will be applicable for every day of delay beyond the specified timeline.		
SLA target	Compliance % = 100%		
Severity level for computation of Liquidated Damages	2		
Assumptions, if any			
Measurement interval	Quarterly		
Reporting interval	Quarterly		
Reports or Data to be submitted, if any	None		
Illustration of LD computation	SLA target	Achievement	LD applicable %
	Compliance % =100% Due date for submission in KD portal is 1 st February 2020	Actual submission done on 20 th February 2020	0.2% corresponding to severity level 2 for missing the compliance target + 0.01%x 19 days of delay beyond due date = 0.2% + 0.19% = 0.39%

1.6.13 To-Be SLA 44 - Compliance to schedule of periodic activities

SLA Description	Compliance to schedule of periodic activities
Scope of SLA	Applies to all periodic IT operations activities described in Volume-II of RFP and information security activities to be carried out by the MSIP as described in Clause 24 of Volume-II – Information Security (Scope of Work). This will include but not be limited to: i. BCP/DR drills ii. Quarterly Security Training and Awareness Exercise iii. Quarterly configuration audit for Attack Surface Reduction iv. Vulnerability assessment v. Quarterly access reconciliation vi. Weekly scan for malware vii. Any other periodic activity described in Volume-II of RFP
Tools, methods or scripts to be used to measure SLA	MSIP should create a detailed calendar of activities and submit to UIDAI at commencement of the contract and update the same on an annual basis or as and when additional activities are added to MSIP's scope for periodic

SLA Description	Compliance to schedule of periodic activities		
	conduct. The same should be signed off by ADG – Operations and ADG – IS of UIDAI.		
Process to capture data or measure the parameter	Submission of respective deliverables upon completion of the scheduled activity and weekly project updates submitted by MSIP to UIDAI.		
SLA calculation	(Number of scheduled activities completed by MSIP during a quarter / Number of scheduled activities during the quarter as per plan) x 100%		
Severity level for computation of Liquidated Damages	SLA %		Severity Level
	100%		None
	<100% and >=98%		1
	<98% and >=96%		2
	<96% and >=94%		3
	<94% and >=92%		4
	<92% and >=90%		5
	<90% and >=85%		6
	<85% and >=80%		7
	<80% and >=75%		8
	<75% and >=70%		9
	<70%		10
Assumptions, if any	NA		
Measurement interval	Quarterly		
Reporting interval	Quarterly		
Reports or Data to be submitted, if any	Submission of respective deliverables upon completion of the scheduled activity		
Illustration of LD computation	SLA target	Achievement	LD applicable %
	Compliance % =100%	94%	0.3%

1.7 Maintenance of Project Documents SLA

1.7.1 To-Be SLA 45 – Maintaining up-to-date project documents

Track / Category	Target	Applied Severity level
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Project documents	Ensure all master project documents are up-to-date, in each Quarter. (Say last date of Quarter completion be A)			
	<=A+15 days	0		
	>A+15 days and <=A+30 days	1		
	>A+30 days and <=A+45 days	2		
	>A+45 days	3		
SLA Description	Maintaining up-to-date project documents.			
Definition of SLA	This SLA will be applicable for update of all project documents, as defined below, on a quarterly basis.			
Scope of SLA	This SLA will be applicable to the following project documents, but not limited to: - SOPs - Architecture document - High level designs and Low-level designs - Integration and Configuration document - Batch processes			
Tool Used for SLA	Not applicable, data to be captured manual			
Data Points for SLA Calculation	Not applicable			
SLA Calculation	Days variance = [(Actual date of update of all project documents during the quarter) – (Last date of quarter)]			
Assumptions	NA			
Measurement Interval	Quarterly			
Reporting Interval	Quarterly			
Reports and data to be submitted	Quarterly SLA Report			
Example	Consider the last date of a particular quarter = 1 st Jan 2020			
	- In Scenario 1, date of update of all project documents = on or before 1st Jan 2020; - In Scenario 2, date of update of all project documents = 8th Jan 2020 - In Scenario 3, date of update of all project documents = 16th Jan 2020 - In Scenario 4, date of update of all project documents = 31st Jan 2020 - In Scenario 5, date of update of all project documents = 5th Feb 2020			
	Scenario	Completion Date	Delay in days	Severity Level
	1	1 st Jan 2020	0-0=0	0
	2	8 th Jan 2020	8-1=7	0
	3	16 th Jan 2020	16-1=15	0
	4	31 st Jan 2020	31-1=30	1
	5	5 th Feb 2020	36-1=35	2

1.8 Key resources SLAs

1.8.1 To-Be SLA 46 – Number of Resource replacements in a Year (for “Key Resources”)

Track / Category	Target	Applied Severity level
Key Resources	Up to ‘1’ replacement	0
	Every increase of ‘1’ replacement	Increases by 2
SLA Description	Replacement of “Key Resources”	
Definition of SLA	Number of resource replacements for “Key Resources” <<RFP clause reference to added>>	
Scope of SLA	This SLA will be applicable to all the “Key Resources”, for whom replacement is initiated by MSIP	
Tool Used for SLA	Not applicable, data to be captured manual	
Data Points for SLA Calculation	Not applicable	
SLA Calculation	Total number of “Key Resource” replacements, which were initiated by the MSIP, in a Quarter Note: The SLA calculation will not include the “key resources” for whom replacement is initiated by UIDAI, instances of resignation, termination of services by MSIP, death, maternity leaves and medical leaves longer than 2 months	
Assumptions	NA	
Measurement Interval	Annual	
Reporting Interval	Annual	
Reports and data to be submitted	Annual SLA Report	

1.9 Technology upgrades SLAs

1.9.1 To-Be SLA 47 – Upgrade of applications to support the (N-1) version of technology components

Track / Category	Target	Applied Severity level
Technology upgrade	100% completion, or UIDAI does not approve the upgrade of any technology components in the Year	0
	>=70% to <100%	1
	< 70% completion or MSIP does not submit the report in the prescribed duration	2
SLA Description	Technology upgrades	
Definition of SLA	This SLA is intended to determine the percentage of all software products and components used to provide IaaS and PaaS services that have been upgraded to support the (N-1) version of technology components in the production and	